

**ScholarPath AdDU: A Web-Based Centralized Scholarship Discovery and Application
Tracking System for Ateneo de Davao University Students**



**A Research Project
Presented to the Faculty of the Computer Studies Cluster School of Arts
and Science Ateneo de Davao University, Davao City**

**In Partial Fulfillment of the Requirements in Capstone Project and Research 1
2nd Semester, SY 2025-2026**

**Operario, Raphael Miguel
Belen, Brendon Justine
Espinosa, Eriel John**

**Adviser:
Adrian Ablazo, MSc.**

April 2026

Table of Contents

Chapter 1: INTRODUCTION

1.1 Background of the Study.....	1
1.2 Problem Statement.....	2
1.3 Objectives of the Study.....	3
1.4 Significance of the Study.....	3
1.5 Scope and Limitations.....	4

Chapter 2: REVIEW OF RELATED WORKS

2.1 Digital Transformation in Higher Education Administration	6
2.2 Web-Based Scholarship Management System.....	6
2.3 Data Organization and Automated Matching in Information Systems.....	7
2.4 Search and Filtering Mechanisms for Financial Aid Discovery.....	7
2.4.1 Policy Desynchronization in Manual Systems.....	8
2.5 Accessibility and Remote Access in Web-Based Systems.....	8
2.6 Usability and User Experience in Educational Platforms	9
2.7 Policy, Data Privacy, and Access Control in Financial Aid Information Systems.....	9
2.8 Theoretical Framework.....	10
2.9 Synthesis of RRL and Research Gaps.....	11

Chapter 3: METHODOLOGY

3.1 Research Design.....	12
3.2 System Development Life Cycle (SDLC).....	13
3.3 System Architecture and Algorithmic Logic.....	15
3.3.1 The "Smart Eligibility Checker"	15
3.3.2 Dynamic Faceted Search and Backend Indexing Strategy.....	18
3.3.3 System Architecture and Cloud Infrastructure.....	18
3.3.4 Role-Based Access Control (RBAC) and Data Privacy Compliance.....	20
3.3.5 Automated Notification Subsystem.....	21
3.4 Testing & Evaluation Procedures: ISO/IEC 25010 Task-Based Usability Protocol.....	21
3.4.1 Step-by-Step Testing Procedure.....	21
3.4.2 Data Analysis and Statistical Tools.....	22
3.4.3 Pre-Study Student Problem Validation Survey and Post-Prototype Perceived Effort Survey	24
3.5 Ethical Considerations.....	26

Chapter 4: THEORETICAL BACKGROUND

4.1 Introduction.....	27
4.2 Relational Database Theory and Normalization.....	27
4.2.1 The Relational Model and Anomaly Prevention.....	27
4.2.2 Progression Through Normal Forms.....	28
4.3 Centralized Information Systems and Cloud Infrastructure Theory.....	28
4.4 Metadata-Driven Document Management.....	29
4.5 Rule-Based Expert Systems Theory.....	29
4.5.1 Knowledge Representation and the Inference Engine.....	29
4.5.2 Conflict Resolution and the Exclusion Flag Hierarchy.....	30
4.6 Information Foraging Theory and Faceted Search.....	30
4.7 Event-Driven Architecture for Asynchronous Notification.....	31
4.8 Role-Based Access Control and Data Privacy.....	32
4.9 Empirical Process Control Theory in Agile Development.....	32
4.10 Human-Computer Interaction and the ISO/IEC 25010 Model.....	33
 Reference.....	 35
 Appendix	
A Financial Aid Pipelines Listed at AdDU.....	39
B Pre-Study Student Problem Validation Survey.....	42
C Post-Prototype Perceived Effort Survey.....	44

Chapter 1: INTRODUCTION

1.1 Background of the Study

Globally, universities are transitioning from traditional, paper-based administrative workflows to digital ecosystems to enhance student access to educational funding. This digitalization is increasingly necessary due to rising educational costs and expanding student populations. In the Philippines, the steady growth of the higher education sector [29] has coincided with tuition fee increases across numerous private institutions, driven by inflation and operational adjustments [11, 19]. While financial aid is available through state-sponsored grants [e.g., 21, 44], international educational investments [45], and private corporate foundations [e.g., 1, 40], discovering and accessing these opportunities remains a challenge due to decentralized information.

At Ateneo de Davao University (AdDU), students face similar financial pressures, exacerbated by recent and proposed tuition increases. To support its student body, AdDU manages a complex financial aid ecosystem comprising 54 distinct funding pipelines, which include internally funded endowments, external corporate partnerships, and state-sponsored grants [2, 8]. However, the application process for these diverse avenues remains largely traditional and decentralized. Students are required to navigate fragmented information sources and submit physical documentation, such as Form 230-SCH and sealed recommendation letters. This reliance on manual documentation creates severe operational bottlenecks for the Office of Student Affairs (OSA) and places an administrative burden on applicants, frequently resulting in eligible students remaining unaware of assistance and scholarship funds being left unutilized.

Table 1: Table of AdDU Financial Aid Programs

Funding Origin / Category	Total Programs	Prime Examples
Internally Funded Endowments	23	Grant-In-Aid (GIA), Jubilee Fund, Jesuit Memorial Endowments
Corporate & External Foundations	17	SM Foundation, Aboitiz Future Leaders, Davao Light (MICD)
State-Sponsored Grants	13	CHED PESFA, DOST-SEI Merit, GSIS Subsidy
Specialized Service Pipelines	1	Student Assistant (SA) Program
Total Validated Pipelines	54	

However, the application process for these 54 distinct financial aid avenues remains fragmented and largely manual, creating systemic inefficiencies that no existing platform at AdDU has resolved.

To address this, this project proposes ScholarPath AdDU — a centralized web-based platform whose primary technical contribution is a rule-based automated eligibility matching engine, the Smart Eligibility Checker, which cross-references a student's Quality Point Index (QPI), household income, and enrolled degree program against the specific, intersecting constraints of all 54 grants simultaneously. Unlike existing scholarship portals in the Philippine higher education literature, which stop at information dissemination or submission routing, the Smart Eligibility Checker employs a forward-chaining inference engine with an Exclusion Flag Hierarchy — a conflict resolution mechanism that ensures specific policy exclusions unconditionally override general eligibility criteria before any grant is surfaced to a student's dashboard. This removes the burden of manual self-assessment that prior systems [14] have identified as the primary driver of erroneous and mismatched scholarship applications. The platform's second core innovation, the Document Vault, employs a metadata-driven, one-time-upload architecture backed by Supabase Storage, allowing students to attach the same verified document to multiple applications without re-submission and enabling OSA administrators to validate documents through a unified, role-governed dashboard.

The third architectural contribution of the platform is a proactive, event-driven notification subsystem that delivers automated SMS and email alerts via Twilio and SendGrid, triggered both by deadline proximity and by application status mutations — ensuring that students receive time-critical updates without relying on passive communication channels that have historically contributed to missed deadlines and unutilized funding at the institutional level.

1.2 Problem Statement

Ateneo de Davao University (AdDU) lacks a centralized digital ecosystem to manage its financial aid landscape of over 50 distinct programs. The university's current reliance on outdated, manual application processes creates a systemic breakdown in the scholarship pipeline, which is particularly urgent given recent tuition fee increases.

Specifically, the current system presents the following problems:

1. **Information Fragmentation and Discovery Friction:** Scholarship details are scattered across 54 distinct internal and external pipelines. Because information is decentralized, students struggle to identify eligible funding and must manually align internal university processing deadlines with unpredictable external corporate timelines.
2. **Inefficient Processing and Matching Errors:** The requirement for physical document submission (e.g., Form 230-SCH) causes severe operational delays at the OSA. Additionally, the lack of automated matching requires students to manually navigate complex eligibility criteria, frequently resulting in application errors and mismatched submissions.
3. **Tracking and Notification Deficits:** The absence of proactive, automated status updates and notifications leads to missed application deadlines, disqualifying students from securing necessary financial aid.

Specific Research Questions

Consequently, this study seeks to answer the following specific questions:

1. What is the perceived reduction in the time AdDU students spend discovering and applying for financial aid when using the centralized ScholarPath AdDU repository compared to the current manual process?
2. How does the Smart Eligibility Checker perform in terms of (a) functional correctness — specifically, whether the engine's Exclusion Flag Hierarchy accurately produces expected outputs across edge-case student profiles as evaluated through a structured test case matrix — and (b) perceived matching effectiveness, as reported by student participants comparing their prior experience of manual eligibility self-assessment against their post-prototype perception of automated eligibility determination?
3. To what extent does the automated notification subsystem reduce the incidence of missed application deadlines among AdDU students?
4. What is the overall system usability and performance efficiency of ScholarPath AdDU as evaluated by users based on the ISO/IEC 25010 functional suitability standards?

1.3 Objectives of the Study

General Objective: To design, develop, and evaluate ScholarPath AdDU, a web-based centralized scholarship discovery and application tracking system for Ateneo de Davao University students. The platform aims to replace manual application processes with a digital system that provides centralized access to financial aid information.

Specific Objectives:

1. To develop an automated eligibility matching engine — the Smart Eligibility Checker — that evaluates a student's QPI, household income, and enrolled degree program against the intersecting constraints of all available financial aid programs, applying a priority-based Exclusion Flag Hierarchy to resolve policy contradictions and surface only verifiably eligible grants to the student's dashboard.
2. To design a centralized digital repository that integrates AdDU's portfolio of over 50 distinct scholarship programs — including internal grants, external corporate partnerships such as Davao Light and SM Foundation, and state-sponsored grants such as CHED and DOST — within a unified, searchable platform supported by compound-indexed dynamic faceted search.
3. To implement a proactive notification and tracking system that alerts students to application statuses and strictly enforced deadlines through automated, event-driven SMS and email delivery.
4. To administer a Pre-Study Student Problem Validation Survey among current or former AdDU scholarship applicants to establish baseline data on the perceived time and effort burden of the existing manual process, and to administer a Post-Prototype Perceived Effort Survey to the same cohort following prototype testing, enabling a direct comparative analysis of perceived effort reduction across Research Questions 1 through 3.

5. To evaluate the ScholarPath AdDU functional prototype using a task-based usability testing protocol aligned with ISO/IEC 25010 standards, measuring functional suitability, performance efficiency, and system usability via the System Usability Scale (SUS).

1.4 Significance of the Study

The implementation of ScholarPath AdDU provides practical, structural benefits to key stakeholders within the university ecosystem. For students, the system offers a streamlined, digital gateway that consolidates financial aid discovery and eliminates the need for manual, physical document submissions. By automating eligibility matching, the platform ensures equitable access to information, helping to mitigate the discovery friction that prevents students from securing funding.

For the Office of Student Affairs (OSA), the platform digitizes application tracking, allowing administrators to verify documents and process approvals through a unified dashboard. This reduces the operational bottlenecks caused by managing fragmented records across 54 funding pipelines, enabling staff to allocate more time to higher-value advisory functions.

Additionally, this study contributes to the broader field of higher education management by providing a documented architectural framework for migrating from paper-based to fully digital scholarship systems. It serves as a replicable model for similar institutions managing multi-pipeline financial aid portfolios under the constraints of national data privacy legislation.

1.5 Scope and Limitations

Scope

This study is bounded by the design, development, and evaluation of a functional web and mobile application prototype for Ateneo de Davao University. The platform's scope is defined by its target user base, the range of financial aid information it consolidates, and the specific administrative functions it digitizes. It does not extend to actual monetary disbursement, integration with the university's finance ledger, or the granting of direct database access to external scholarship organizations. Within these boundaries, the system addresses the full digital lifecycle of scholarship discovery and application tracking — from initial student profile registration and automated eligibility matching through to document submission, deadline notification, and OSA-level application processing.

The system scope includes:

- **Target Audience Coverage:** The platform is designed exclusively for the undergraduate college-level student body, authorized administrators within the AdDU Office of Student Affairs (OSA), and designated academic administrators, specifically Department Chairs and Coordinators.

- **Information Coverage:** The centralized repository integrates eligibility criteria, deadline schedules, and benefits for over 50 financial aid programs. This includes government grants (e.g., CHED, DOST), corporate endowments (e.g., Aboitiz, SM Foundation), and AdDU internal programs (e.g., Grant-in-Aid, Working Student grants).
- **Core Functionality:**
 - An automated algorithm that evaluates student profiles (Quality Point Index, household income, and enrolled degree program) to provide personalized scholarship matching.
 - A secure digital repository allowing students to upload necessary prerequisites (e.g., academic transcripts, BIR-stamped ITRs) once to attach to multiple applications, eliminating the need for manual physical submissions.
 - An automated notification subsystem delivering critical deadline alerts and status updates via SMS and email.
 - A backend Content Management System (CMS) enabling OSA administrators to broadcast announcements, monitor pending requests, and track real-time analytics. Additionally, the system features a dedicated review interface for Department Chairs and Coordinators to efficiently filter and assess applicant economic statuses for Grant-in-Aid (GIA) selections.
- **Development Approach:** The prototype implements a hybrid mobile and web architecture to ensure cross-platform compatibility. The system utilizes standard web frameworks and cloud-based databases to manage real-time synchronization, cloud storage, and secure user authentication.
- **Evaluation Scope:** Usability testing based on ISO/IEC 25010 standards will involve a purposive sample of approximately 10 to 15 AdDU college students, active OSA administrative staff, and selected Department Chairs or Coordinators. The evaluation will use a mixed-method approach, combining task-based observational testing (measuring completion rates and time-on-task) with two sequential post-session instruments: the System Usability Scale (SUS) to assess perceived interface usability, and the Post-Prototype Perceived Effort Survey (Appendix C) to generate comparative data against the pre-study baseline from Appendix B. Results will be quantified using descriptive statistics to measure the perceived reduction in processing times.

Limitations

This study acknowledges the following realistic constraints that define its boundaries:

- **Financial Transaction Limitation:** The prototype functions solely as a discovery, matching, and application-tracking platform. It does not handle actual monetary disbursements, bank routing, or direct tuition fee deductions from the university's finance ledger.
- **Administrative Access Constraints:** The system centralizes information regarding external scholarships but does not grant external organizations (e.g., DOST, SM Foundation) direct administrative access to the AdDU database.

- **External Submission Dependencies:** For specific external grants, the system acts only as an information and tracking hub. Students may still be mandated to complete final submissions or examinations through official third-party portals, such as the DOST E-Scholarship Application System.
- **Predictive Matching Only:** The automated matching engine predicts eligibility based on the input data; it does not guarantee a grant. The final evaluation, interview assessments, and approval of financial aid remain under the strict, manual jurisdiction of the AdDU College Scholarship Committee, Department Chairs and Coordinators (for GIA), or the external granting agencies.
- **Registrar Integration Limitation:** The platform does not integrate with the AdDU University Registrar's information system. Student QPI data is self-reported by the student during account registration and is stored exclusively within ScholarPath AdDU's PostgreSQL database. This data is not cross-referenced against or validated against the Registrar's official academic records prior to eligibility matching. Consequently, the accuracy of eligibility determinations is contingent on the honesty and accuracy of the student's self-reported academic standing.
- **Budget Allocation Constraint:** The platform does not implement an algorithm for determining which among multiple qualified applicants should receive a scholarship award when the number of eligible applicants exceeds the number of available funded slots for a given grant. The system's automated matching engine identifies and surfaces all students who meet a scholarship's eligibility criteria; however, the final awarding decision — including any priority ranking or selection among equally qualified applicants — remains under the exclusive jurisdiction of the AdDU College Scholarship Committee, the relevant Department Chairs and Coordinators (for Grant-in-Aid selections), or the respective external granting agencies. A future system enhancement could explore weighted scoring or preference-based allocation algorithms to assist administrators in this selection phase, but such functionality is explicitly outside the scope of the present prototype.
- **User Base Limitation:** System access is restricted exclusively to bona fide, officially enrolled AdDU college undergraduate students and specific university personnel. Basic education units (Grade School, Junior High, Senior High), Graduate School students, and the general public are excluded from this platform.
- **Testing Scope Limitation:** Usability evaluation involves a purposive sample of participants rather than a university-wide deployment. Findings will inform prototype refinement for the capstone requirements but do not constitute definitive statistical claims about the system's efficacy across the entire AdDU population.
- **Prototype-Based Time Estimation Constraint:** The participant time estimates collected through the Post-Prototype Perceived Effort Survey (Appendix C) are based on a single, controlled prototype testing session rather than sustained real-world use of the deployed system. Consequently, these estimates may not fully reflect actual time-on-process in practice, as factors such as varying network conditions, individual document readiness, and accumulated platform familiarity over repeated use could produce meaningfully different results from those observed during the evaluation session.

Chapter 2: REVIEW OF RELATED WORKS

2.1 Digital Transformation in Higher Education Administration

The transition from traditional administrative processes to digital ecosystems is a critical focus for modern educational institutions. Early Management Information Systems (MIS) successfully digitized basic student records but frequently functioned as siloed databases that failed to integrate cross-departmental workflows [3, 5]. This fragmentation resulted in underutilized resources, compromised administrative efficiency, and unequal educational access — not merely because paper was replaced by screens, but because the underlying workflows remained disconnected. Institutions globally have recognized that streamlining student services requires unified digital workflows as a structural prerequisite for operational sustainability [46].

Contemporary digital transformation therefore requires interoperable architectures — systems capable of unifying data collection, processing, and reporting across departments into a single, coherent workflow [36]. Kayanja et al. [46] confirm in their 2025 study of fully automated and paperless systems in higher education that the most significant efficiency gains emerge not from digitizing isolated records, but from architecturally integrating those records into cross-departmental process flows. This finding is directly applicable to scholarship administration, where eligibility data, document submissions, and deadline tracking currently exist as separate, disconnected processes at most Philippine higher education institutions. The literature therefore establishes that the effectiveness of digital transformation in higher education depends on the adoption of interoperable, cross-departmental architectures — the foundational design principle motivating ScholarPath AdDU's centralized, Supabase-backed infrastructure.

2.2 Web-Based Scholarship Management Systems

While web technologies are increasingly leveraged for scholarship management globally, the literature focusing on Southeast Asian and Philippine higher education contexts reveals both what existing systems achieve and where they fall critically short. Understanding what these systems implement is necessary before identifying the gap that ScholarPath AdDU is designed to fill.

Bagaporo and Espinosa [12] developed an Online Scholarship Management System for Camarines Sur Polytechnic Colleges (CSPC), implementing a centralized digital repository that consolidates scholarship listings, automates announcement dissemination to enrolled students, and provides an administrative dashboard for monitoring scholarship status. The system successfully eliminated the need for physical bulletin boards and improved information reach across the multi-campus institution. However, the platform stops at information dissemination: document submission remained a manual, offline process, and there was no mechanism for automated eligibility matching. Students who discovered a scholarship through the portal were still required to process physical paperwork and submit to the respective offices in person [12].

Daluyon and Bilog [14] developed a Scholarship Management Information System specifically for State Universities and Colleges (SUCs) in Rizal, incorporating online application submission and a structured administrative review interface. Their system advanced beyond the CSPC implementation by digitizing the submission process, reducing paper routing, and enabling administrators to view and sort applications from a centralized dashboard. Crucially, however, Bilog's evaluation found that the absence of automated eligibility validation was the primary driver of application errors and administrative rework across multi-campus deployments. Students continued to self-assess their eligibility and frequently submitted applications for grants they did not qualify for, generating review bottlenecks for administrative staff [14]. This finding directly motivates the automated Exclusion Flag Hierarchy in ScholarPath AdDU's Smart Eligibility Checker.

Al-Ayyubi and Maulana [3] designed a web-based scholarship management information system for a higher education institution in Indonesia, achieving efficient dissemination of scholarship information and basic application form routing through a unified web interface. The system demonstrated measurable improvements in processing speed and information accessibility. Yet the researchers themselves identified the architectural absence of integrated, secure document handling as a critical and recurring gap — applicants uploaded details through forms, but prerequisite documents such as transcripts and income proofs were still submitted physically and managed separately [3]. This precise gap informs the Document Vault architecture in ScholarPath AdDU, which is designed to decouple document storage from application routing and enable one-time upload, multi-attach reuse.

At the international level, Amer [5] at the Lebanese American University developed a Smart Scholarship Management System as an all-in-one solution, integrating application submission, document handling, and status tracking into a unified platform. The system demonstrated the technical feasibility of end-to-end digital scholarship administration but was built for a single institution with a small, well-defined portfolio of scholarships and did not address multi-variable eligibility matching across taxonomically diverse grant categories. The localization challenge — adapting to the intersection of internal endowments, external corporate foundations, and state-sponsored grants under distinct eligibility taxonomies — was not a design constraint of that system.

The reviewed literature therefore reveals a clear progression in capability: from static digital repositories [12], to submission-enabled portals [14, 3], to integrated multi-function platforms [5] — with none demonstrating a validated, end-to-end solution that simultaneously integrates automated multi-variable eligibility matching, secure document vaulting, and proactive notification within a single platform tailored to a localized university taxonomy. This convergent architectural gap directly motivates the ScholarPath AdDU system.

2.3 Data Organization and Automated Matching in Information Systems

In digital scholarship repositories, the technical mechanism by which students are matched to eligible grants varies substantially across the reviewed literature — and these differences in mechanism, not merely in feature completeness, determine the quality of matching outcomes.

Three dominant implementation patterns emerge: keyword-overlap matching, threshold-based sequential rule evaluation, and machine-learning-driven classification, each carrying distinct computational behaviors and institutional tradeoffs.

Keyword-overlap matching operates by tokenizing a student's profile fields and tallying term coincidences against grant descriptions. Systems relying solely on this mechanism are computationally incapable of evaluating intersecting numerical constraints — QPI thresholds, household income ceilings, degree program restrictions — as a unified logical gate. Orgianus et al. [36] document this explicitly: keyword-based retrieval produced persistent false-positive matches because a keyword engine performs term frequency comparisons, not constraint satisfaction, and cannot evaluate "QPI $\geq$ 2.75 AND income $\leq$ ₱300,000 AND degree $\in$ {BS CS, BS IT}" as a single pass/fail expression.

Threshold-based sequential rule evaluation, as implemented by Bilog [14], advances beyond keyword matching by encoding eligibility criteria as conditional logic — but evaluates constraints independently and without formal priority ordering, producing conflict resolution failures at edge cases that Section 2.5 examines in detail.

A third pattern — machine-learning classification — is documented by Kanwal et al. [50], whose C4.5 decision tree achieved 95.62% prediction accuracy on scholarship eligibility data. However, ML-derived decision boundaries carry two institutional liabilities for a multi-pipeline environment like AdDU: the model requires retraining whenever grant criteria change, and its decisions offer no interpretable audit trail — a compliance requirement under RA 10173 that the authors themselves acknowledge [50].

These findings establish a technical hierarchy of tradeoffs: keyword matching fails at constraint evaluation; sequential rule evaluation fails at conflict resolution; ML classification fails at policy interpretability and update responsiveness. ScholarPath AdDU's Smart Eligibility Checker addresses all three through a forward-chaining rule engine with an Exclusion Flag Hierarchy, the design specifics of which are examined in the context of prior deployed implementations in Section 2.5.

2.4 Search and Filtering Mechanisms for Financial Aid Discovery

To understand why dynamic faceted search represents a technically superior retrieval architecture for scholarship repositories, it is necessary to examine how keyword search operates at a structural level — not merely what it fails to do, but why it fails given the specific retrieval demands of a multi-pipeline financial aid database.

Keyword search systems maintain, at their technical core, an inverted index: a data structure that maps individual terms to the ordered list of records in which they appear (posting lists). When a user submits a query, the engine retrieves the posting lists for each query term and intersects or unions them, ranking results by term frequency or relevance weighting [43]. This mechanism performs correctly when query vocabulary aligns with document vocabulary — but scholarship repositories present a structural retrieval mismatch on both sides of this requirement. On the query

side, students are frequently unaware of the precise grant name, sponsor organization, or category keyword: a student searching "income-based scholarship" retrieves zero results from a grant catalogued as "Grant-in-Aid Fund," despite the GIA being precisely what that student qualifies for. On the document side, a keyword engine is structurally incapable of filtering by continuous numerical attributes — it cannot natively express "show me grants where a QPI of 2.8 satisfies the minimum threshold" without a separate constraint-evaluation layer, because inverted indexes store term occurrences, not numerical bounds. Mahdi et al. [4] confirm this limitation in their review of 47 deployed information retrieval systems, finding that keyword-only retrieval consistently underperforms in repositories where items carry multiple intersecting categorical and numerical attributes.

Dynamic faceted search resolves this structurally by inverting the retrieval model: rather than the user constructing a query and the system returning documents, the system exposes its underlying metadata taxonomy as interactive filter controls, allowing users to iteratively reduce a large result set by selecting attribute values — without requiring prior knowledge of specific grant names or internal classification conventions [43]. The query mechanism shifts from posting-list intersection over terms to predicate conjunction over structured metadata fields, a fundamentally different computational operation that directly supports the multi-attribute nature of scholarship eligibility.

The performance characteristics of this approach, however, are tightly coupled to the indexing architecture underlying the query executor. For categorical attributes with low cardinality — such as grant category (four values in ScholarPath AdDU's taxonomy) or active status (binary) — B-Tree indexes support $O(\log n)$ key lookups and range scans that retrieve qualifying rows without full-table sequential scans. For continuous numerical attributes — specifically QPI requirements and application deadlines — Range indexes support boundary-constrained retrieval, enabling the query planner to access directly the subset of records satisfying $\text{Grant.MinimumQPI} \leq \text{student.QPI}$ without scanning disqualified rows [36]. When compound indexes are constructed across combinations of these fields — for example, a composite index on (Grant_Category, Status, QPI_Requirement) — the query planner can satisfy multi-facet filter combinations from a single traversal rather than executing independent scans per facet and joining their results in memory, reducing query execution complexity from $O(k \times n)$ to $O(\log n)$ for k simultaneously active filters. Tunkelang [43] establishes that the quality of the underlying metadata taxonomy is the decisive determinant of whether this architecture reduces or merely redistributes cognitive load: poorly normalized facets produce overlapping filter categories that reintroduce the ambiguity the search was designed to eliminate.

What the reviewed literature does not document is a validated deployment of this compound-indexed faceted search architecture within a Philippine higher education financial aid portal managing the concurrent taxonomic diversity of internal endowments, external corporate foundations, and state-sponsored grants — the precise multi-source structure of AdDU's 54-pipeline portfolio. This is not a feature gap but an architectural and taxonomic gap: prior implementations either deploy keyword retrieval over a homogeneous grant set, or deploy faceted search over repositories without the compound categorical-and-numerical attribute structure that characterizes AdDU's financial aid taxonomy. ScholarPath AdDU addresses this through B-Tree and Range

compound indexes over four primary metadata facets, with client-side input debouncing at 300 milliseconds to prevent redundant query execution during iterative filter selection.

2.4.1 Policy Desynchronization in Manual Systems

A critical vulnerability of decentralized, paper-based administrative systems is policy desynchronization — the structural mismatch between government fiscal year release cycles and academic year timelines [16], compounded by the multi-campus application error patterns documented by Bilog [14]. When external partners update their eligibility criteria or deadlines, localized bulletin boards — maintained as static HTML pages or physical notices — have no programmatic mechanism to propagate these changes to dependent application records or notify students of the update. The data model underlying these systems is the root cause: eligibility information and deadline data are stored as unstructured content in page-rendering layers rather than as structured, queryable records in a relational database. A change to a grant's QPI threshold is therefore a content editing operation, not a data update, and carries no guarantee of consistency across all instances where that criterion is referenced. Catibog [16] identifies this structural weakness as a primary driver of the discrepancies between student-facing scholarship criteria and the criteria actually applied by administrative offices at award time. Centralized, database-driven platforms are architecturally positioned to resolve this through a single-source-of-truth model — but validated implementations of such systems that explicitly manage the interplay between internal university processing timelines and unpredictable external portal schedules remain absent from the reviewed Philippine HEI literature.

2.5 Rule-Based Matching and Faceted Search in Deployed Systems

While Section 2.3 established the theoretical case for automated eligibility matching and Section 2.4 identified dynamic faceted search as the leading discovery mechanism, it is necessary to examine how these technologies have been implemented in actual deployed systems. This examination provides the empirical grounding for the algorithmic choices made in ScholarPath AdDU and distinguishes the present study's implementation from prior work.

Rule-based eligibility matching has been deployed in several educational information systems, with varying degrees of constraint complexity. In the Philippine context, Bilog [14] implemented a rule-based validation layer within the Scholarship Management Information System for SUCs in Rizal, where eligibility constraints — including GPA thresholds, institutional affiliation, and enrollment status — were encoded as conditional logic applied during the application intake process. While this implementation successfully reduced erroneous submissions at the point of entry, it evaluated constraints sequentially and independently, without a mechanism for resolving conflicts when multiple rules applied simultaneously. This limitation — the absence of a priority-based conflict resolution strategy — resulted in edge cases where students holding concurrent government grants passed initial validation for additional state-sponsored awards [14]. ScholarPath AdDU addresses this directly through the Exclusion Flag Hierarchy, a priority-based conflict resolution mechanism in which specific negative constraints unconditionally override general

positive inclusion rules, a design choice derived from the conflict resolution gap identified in Bilog's implementation.

At the international level, Amer [5] implemented an eligibility filtering component within the Smart Scholarship Management System at the Lebanese American University that matched student academic profiles against scholarship criteria defined in a centralized database. The system used a multi-criteria filter that evaluated grade point average, program of study, and nationality. However, the filter operated on a flat, non-hierarchical rule set without weighted prioritization, and the scholarship portfolio managed was taxonomically uniform — all grants originated from the same institutional source, eliminating the policy desynchronization challenge posed by multi-source, externally governed grants [5]. ScholarPath AdDU's matching engine operates across four taxonomically distinct grant categories — internally funded endowments, corporate foundations, state-sponsored grants, and specialized service pipelines — requiring a more granular constraint architecture than prior implementations demonstrate.

Regarding faceted search, Mahdi et al. [4] conducted a comprehensive review of dynamic faceted search implementations across information retrieval systems, identifying compound indexing and real-time metadata synchronization as the dominant performance optimization strategies in large-scale deployments. Their analysis of 47 systems confirms that faceted search outperforms keyword-only retrieval particularly in repositories where items carry multiple, intersecting categorical attributes — precisely the structure of AdDU's scholarship database, where each grant carries attributes across funding origin, coverage type, eligible degree programs, QPI thresholds, and income ceilings [4]. Tunkelang [43] further establishes that the effectiveness of faceted search is contingent on the quality of the underlying metadata taxonomy: poorly normalized facets produce ambiguous or overlapping filter categories that reintroduce the cognitive burden the search is meant to reduce. This finding directly informed ScholarPath AdDU's backend categorization strategy, in which the 54 financial aid pipelines are normalized into four primary metadata facets with compound B-Tree and Range indexes to ensure unambiguous filter behavior and real-time retrieval without system latency.

Taken together, the reviewed implementations establish that rule-based eligibility matching and dynamic faceted search are validated approaches in educational portal contexts, but that no prior system has simultaneously deployed conflict-resolving rule-based matching and compound-indexed faceted search within a single platform managing a taxonomically diverse, multi-source scholarship portfolio. This implementation gap — not the theoretical gap — is what ScholarPath AdDU is positioned to address.

2.6 Accessibility and Remote Access in Web-Based Systems

Remote access in web-based academic systems is not merely a deployment preference — it is an architectural decision that determines whether a system can sustain adoption in mobile-dominant, bandwidth-variable environments such as Mindanao's higher education landscape. The technical architecture a system adopts for cross-platform delivery determines its responsiveness under

variable connectivity, its ability to deliver proactive notifications, and its maintenance overhead — all of which have direct consequences for adoption rates and operational sustainability.

Three dominant architectural patterns exist for cross-platform academic portals, each representing a distinct tradeoff between development complexity, performance characteristics, and native capability access. Responsive web applications (RWAs) adapt a single web codebase to varying viewport sizes through CSS media queries and fluid grid layouts, requiring no separate build for mobile targets but limited to capabilities available through the browser's JavaScript API surface. Progressive Web Applications (PWAs) extend this pattern by registering Service Workers — background scripts that cache application shells and API responses in the browser's Cache Storage — enabling partial offline functionality and reducing cold-start load times under intermittent connectivity [33]. However, iOS imposes persistent platform restrictions on PWA background execution and push notification delivery through WebKit that reduce their reliability as proactive alert channels; scholarship applications with time-critical deadline notifications cannot depend on this model for consistent delivery. Hybrid mobile architectures — specifically the Capacitor framework used in ScholarPath AdDU — resolve this by compiling the same Vite production JavaScript bundle into native Android and iOS packages: Capacitor instantiates a WKWebView (iOS) or WebView (Android) host process and bridges a set of native plugin APIs — including push notifications via APNs and FCM, local filesystem access, and camera — into the JavaScript execution context through a typed bridge interface. This architecture delivers the development efficiency of a unified JavaScript codebase while accessing native OS capabilities that browser-based delivery cannot reach, without requiring a parallel Swift or Kotlin codebase.

The functional case for this architectural choice is reinforced by documented adoption evidence specific to the Philippine context. The World Bank's [48] assessment of digital transformation in Philippine higher education explicitly identifies mobile-first design as a structural prerequisite for adoption — not a secondary usability consideration — particularly for Mindanao-based institutions where smartphone access substantially exceeds laptop ownership rates. Yoo and Huang [49] further establish that platforms perceived as culturally or technologically misaligned with users' existing digital habits generate adoption failure through reversion: when a portal's interface model conflicts with how a user's primary device (typically a mid-range Android smartphone) renders and interacts with web content, users abandon the digital channel and return to manual processes, recreating precisely the administrative bottlenecks the platform was deployed to eliminate.

The reviewed literature therefore establishes both the technical case for a hybrid architecture and the documented adoption consequences of under-engineering the mobile delivery layer. What prior scholarship portal implementations in the Philippine literature have not documented is a formal operationalization of this pattern — specifically the Capacitor-wrapped production build approach producing a single deployable codebase for web, Android, and iOS — within a financial aid management context. ScholarPath AdDU's hybrid architecture directly addresses this gap, enabling the automated notification subsystem to deliver time-critical deadline alerts through native push channels that browser-based web portals cannot reliably access.

2.7 Usability and User Experience in Educational Platforms

In evaluating web-based academic systems, usability testing is universally identified as the most vital assessment method. Task-based usability testing directly measures how well users can complete real tasks under realistic conditions. To complement observed performance, researchers rely on standardized perception-based instruments like the System Usability Scale (SUS). Bangor, Kortum, and Miller [13] conducted an extensive empirical evaluation of the SUS, confirming its high reliability and validity across diverse digital interfaces. Empirical research by Tullis and Stetson [42] further demonstrated that a sample size of 10 to 12 participants is sufficient to yield reliable SUS scores that accurately reflect the broader user experience.

When these standardized instruments are used alongside the ISO/IEC 25010 Software Quality Evaluation model [31], developers can effectively quantify perceived usability and ensure the interface minimizes administrative friction. The evaluative methodology of this study is built on this established consensus: task-based usability testing paired with the SUS and the ISO/IEC 25010 quality framework constitutes the field's most validated approach for determining whether an academic platform genuinely reduces the friction it was designed to address [13, 31, 42]. Applying this protocol to ScholarPath AdDU provides the empirical backbone through which the study's claims about discovery efficiency, eligibility accuracy, and deadline management are substantiated.

2.8 Policy, Data Privacy, and Access Control in Financial Aid Information Systems

Financial aid management systems process data that is simultaneously sensitive, multi-tenanted, and multi-role: a student's household income record must be readable by OSA administrators but invisible to other students; a document's verification status must be writable by OSA staff but read-only to Department Chairs; application records must be queryable by administrators globally but accessible to students only for their own submissions. The technical mechanism by which these constraints are enforced — not merely the policy that states they should be enforced — determines whether a system achieves genuine RA 10173 compliance or only nominal compliance.

The two dominant access control paradigms in information systems differ fundamentally in their enforcement architecture. Access Control Lists (ACLs) enforce access by maintaining, for each data object, an explicit list of subjects and their permitted operations. In a university scholarship context, this means maintaining per-student permission entries for each application record — a model that scales as $O(\text{users} \times \text{objects})$ and requires manual permission reassignment each time a student's status changes or an administrator's scope is modified. More consequentially, ACL enforcement in web applications is typically implemented at the application layer: a middleware check verifies the requesting user's permission before executing the database query. This model carries a structural security vulnerability: if an API endpoint is inadvertently exposed through a misconfigured route, a CORS policy error, a parameter injection attack, or a client-side logic bypass, the database itself executes the query without resistance — the application-layer permission check that was bypassed leaves no second line of defense at the data tier [38].

Role-Based Access Control (RBAC), formalized by Ferraiolo and Kuhn [24] and standardized through NIST [35], resolves the scalability problem through role indirection: permissions are attached to roles rather than to individual users, reducing permission management from $O(\text{users} \times$

objects) to $O(\text{roles} \times \text{permissions})$. The more architecturally significant property of RBAC for RA 10173 compliance, however, is its capacity for enforcement at the database engine level through PostgreSQL's Row Level Security (RLS) mechanism. An RLS policy is a predicate expression — for example, `USING (student_id = auth.uid())` on the Applications table — that PostgreSQL evaluates as part of every query's execution plan, appended automatically before any rows are returned to the requesting process. Because this predicate is evaluated within the query executor itself, it operates below the application layer entirely: there is no code path — whether through a frontend routing error, a misconfigured API proxy, or a direct database connection — through which a student can access another student's application record, because the filter is enforced by the database engine, not by the application that queries it.

Despite this technically superior enforcement model, Halili [26] documents that Philippine HEI scholarship portals commonly lack transparent, auditable access control implementations, and CHED's memorandum guidelines [18] do not specify the technical mechanism through which access segregation must be achieved — leaving institutions free to implement application-layer checks that carry the structural vulnerability described above. The consequence is that existing Philippine scholarship portals may be formally RA 10173-compliant at the policy documentation level while remaining technically vulnerable at the implementation level — a gap that is not visible in policy audits but would be exploitable through standard API testing. ScholarPath AdDU addresses this gap by enforcing its RBAC permission matrix through PostgreSQL RLS policies that bind role-to-permission rules directly to the database engine across all three system roles: undergraduate students (self-access only), OSA administrators (global read/write for application processing), and Department Chairs (read-only, department-scoped). This approach ensures that access segregation is architecturally unavoidable rather than policy-dependent, and that the platform's RA 10173 compliance is verifiable at the database schema level rather than contingent on the correctness of application-layer code.

2.9 Theoretical Framework

This study is anchored on the Information Systems Success Model [20], a widely validated framework that evaluates the effectiveness of centralized web portals. The model posits that the success of an information system is driven by six interrelated dimensions: System Quality, Information Quality, Service Quality, System Use, User Satisfaction, and Net Benefits. As illustrated in Figure 1, this standard model has been explicitly adapted to map the operational variables and institutional constraints of the ScholarPath AdDU platform.

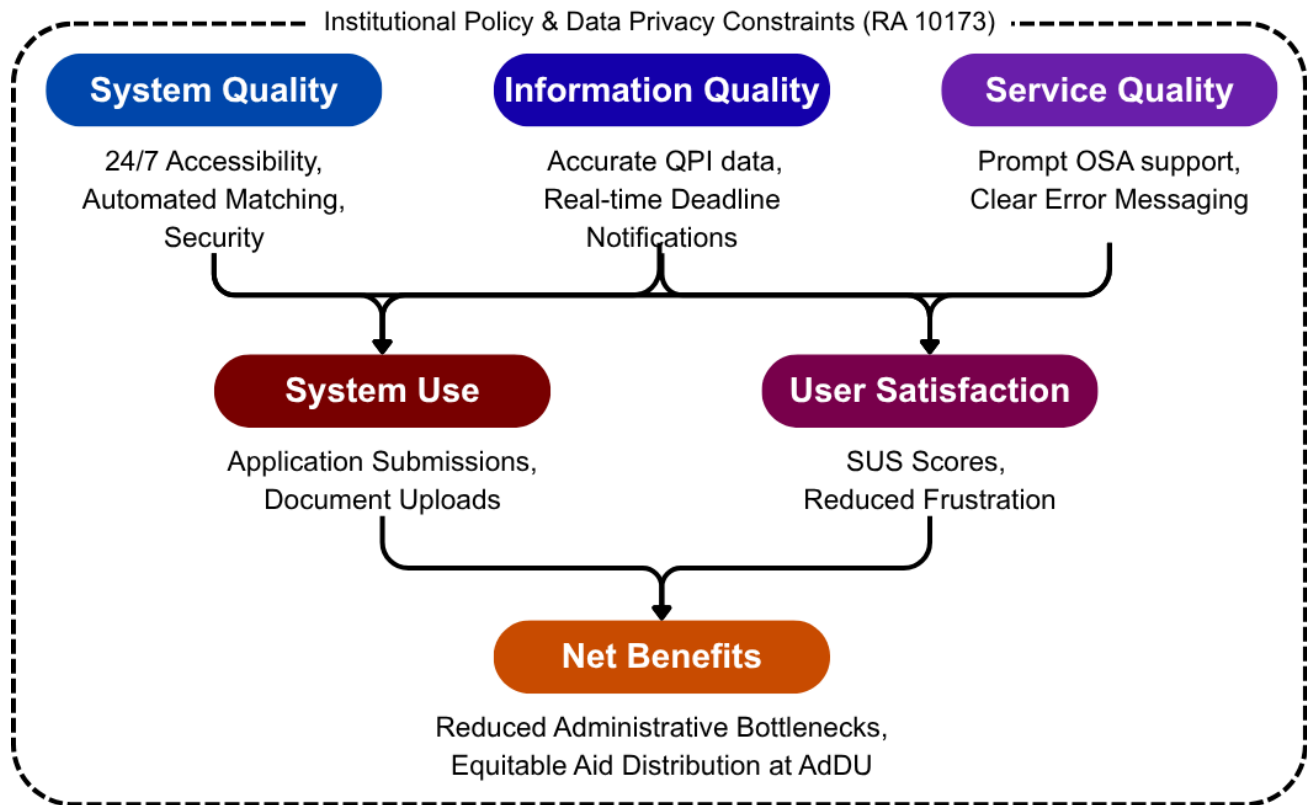


Figure 1: *Adapted Information Systems Success Model for ScholarPath AdDU*

In the context of this study, the foundational dimensions are defined by the platform’s technical and informational reliability. System Quality is represented by the portal’s 24/7 accessibility, the precision of its automated matching algorithms, and its secure data architecture — each of which responds to the system-architecture gaps documented in the reviewed literature [3, 12, 14]. Information Quality is reflected in the accuracy of the consolidated scholarship details, real-time deadline notifications, and precise QPI tracking — addressing the policy desynchronization and information fragmentation identified in [16] and [36]. Service Quality encompasses the system’s responsiveness, including clear error messaging for students and prompt workflow support for the Office of Student Affairs (OSA) — informed by the cultural alignment requirements documented in [49] and [48].

According to the model, high performance in these foundational areas directly drives user engagement. As students and OSA administrators experience reliable system and information quality, their System Use (e.g., uploading prerequisite documents, processing applications) and User Satisfaction (measured quantitatively via the System Usability Scale) will increase. Ultimately, this sustained engagement leads to the system’s Net Benefits: eliminating manual administrative bottlenecks, preventing missed scholarship deadlines, and ensuring an equitable, data-driven distribution of financial aid across the university. Crucially, this entire framework operates strictly within the parameters of institutional policy and national data privacy laws, with all dimensions of system success governed by the compliance requirements of RA 10173 through the RBAC architecture established in [24, 38].

2.10 Synthesis of RRL and Research Gaps

The reviewed literature, taken as a whole, establishes a clear progression of scholarship management system capability — from information dissemination [12] to submission-enabled portals [3, 14] to multi-function integrated platforms [5] — while consistently revealing that no existing implementation simultaneously resolves the full set of architectural challenges posed by a complex, multi-source financial aid environment. The specific technology and system-level gaps motivating ScholarPath AdDU are summarized in Table 2 below, with each gap explicitly traced to the reviewed work that documented it.

Table 2: Synthesis of System Gaps and Proposed Design Responses

Reviewed Work	System/Technology Gap Identified	ScholarPath AdDU Design Response
Al-Ayyubi & Maulana [3]; Bagaporo & Espinosa [12]	Existing scholarship portals stop at information dissemination; prerequisite document submission remains physical and disconnected from the digital workflow.	Document Vault: one-time upload, metadata-tagged, multi-attach architecture using Supabase Storage with foreign key linking to the Applications table.
Bilog [14]	Absence of automated eligibility validation is the primary driver of application errors. Rule-based filtering was implemented but lacked conflict resolution for overlapping constraints.	Smart Eligibility Checker with Exclusion Flag Hierarchy: specific negative constraints unconditionally override general positive inclusions, resolving policy contradictions before surfacing results.
Mahdi et al. [4]; Tunkelang [43]	Basic keyword search fails in large, categorically diverse repositories. Faceted search requires a well-normalized metadata taxonomy and compound indexing for real-time retrieval.	Dynamic Faceted Search with B-Tree and Range compound indexing over four primary metadata facets; client-side debounce (300ms) to minimize server load.
Catibog [16]; Orgianus et al. [36]	Policy desynchronization between government fiscal cycles and institutional academic timelines creates deadline mismatches in decentralized systems.	Centralized Supabase/PostgreSQL backend: any update to a scholarship deadline or criterion propagates instantly across all 54 pipelines and all concurrent user sessions.

Yoo & Huang [49]; World Bank [48]	Culturally misaligned interfaces drive users back to manual processes; mobile-first design is a prerequisite for adoption in Philippine HEI contexts.	Hybrid web and mobile architecture via Capacitor wrapping the Vite/React production build into native Android and iOS packages from a single codebase.
Halili [26]; CHED [18]	Philippine HEI scholarship portals have limited documentation of RBAC operationalization in compliance with RA 10173 in multi-role administrative workflows.	RBAC Permission Matrix enforced at the database engine level via PostgreSQL Row Level Security (RLS), with immutable role-to-permission mapping for students, OSA administrators, and Department Chairs.

The convergence of these gaps — the absence of integrated document workflows, unresolved policy desynchronization, culturally misaligned interfaces, and undocumented RBAC compliance in Philippine HEI contexts — collectively identifies a distinct and unaddressed design space. No existing implementation in the reviewed literature simultaneously integrates multi-variable automated eligibility matching with conflict resolution, a unified digital document vault, compound-indexed faceted search, and a proactive notification subsystem within a single platform tailored to a localized university taxonomy. This gap directly motivates the present study and grounds each architectural decision in ScholarPath AdDU in specific, documented evidence from the reviewed literature.

Chapter 3: METHODOLOGY

This chapter details the systematic procedures, processes, and tools utilized in the design, development, and evaluation of the ScholarPath AdDU system. It outlines the research design, system development methodology, system architecture, data gathering instruments, and statistical tools used for data analysis.

3.1 Research Design

The research design incorporates a comparative usability evaluation to validate the system's impact on processing efficiency. During the initial requirements gathering phase, the researchers will administer a pre-study time-on-task survey to a purposive sample of existing scholars. This survey aims to measure the average duration students currently spend discovering financial aid options and routing physical documents (e.g., Form 230-SCH).

Once the ScholarPath AdDU prototype is developed, a controlled, task-based usability testing session will be conducted. The perceived time and effort recorded in the Pre-Study Student Problem Validation Survey (Appendix B) will then be compared against the corresponding items in

the Post-Prototype Perceived Effort Survey (Appendix C), which is administered to the same participants immediately following the prototype session. Both instruments use identical ordinal and Likert-scaled items, enabling direct item-to-item mean comparisons to determine whether participants perceive the digital workflow to require significantly less time and effort than the manual process they described at baseline. The design and administration of both instruments, and the statistical procedure for their comparison, are detailed in Section 3.4.3. This comparative analysis will quantify the reduction in application processing time and validate the system's perceived performance efficiency.

The comparative evaluation follows a within-subjects design, meaning the same cohort of participants who complete the Student Problem Validation Survey (establishing perceived baseline effort) will subsequently participate in the prototype usability testing session. This design is chosen over a between-subjects approach because it eliminates inter-group variability—each participant serves as their own control, ensuring that any observed difference in perceived effort is attributable to the system rather than individual differences in prior scholarship experience. A sample of 10 to 15 participants is sufficient for this design given that SUS reliability stabilizes at that range [42], and the time-perception comparison uses ordinal scaled data that does not require large-N inferential statistics.

This study employs a Developmental Research Design, specifically focusing on System Development. This approach is the most appropriate for IT and Computer Studies capstone projects, as it centers on the structured creation, implementation, and evaluation of a functional software solution to address a specific operational problem. In this context, the design governs the transition of the AdDU Office of Student Affairs' (OSA) manual scholarship application process into a centralized, web-based digital ecosystem. The developmental design is complemented by a descriptive evaluation phase to assess the prototype's usability and functional suitability.

3.2 System Development Life Cycle (SDLC)

To ensure flexibility and iterative improvement, the development of ScholarPath AdDU follows the Agile Development Methodology. Unlike rigid, linear models, Agile allows the development team to adapt to evolving requirements from stakeholders — including OSA staff, Department Chairs, and Admissions personnel — through continuous feedback loops between phases. As illustrated in Figure 2, the development process is organized into five iterative phases, each building upon the outputs of the last.

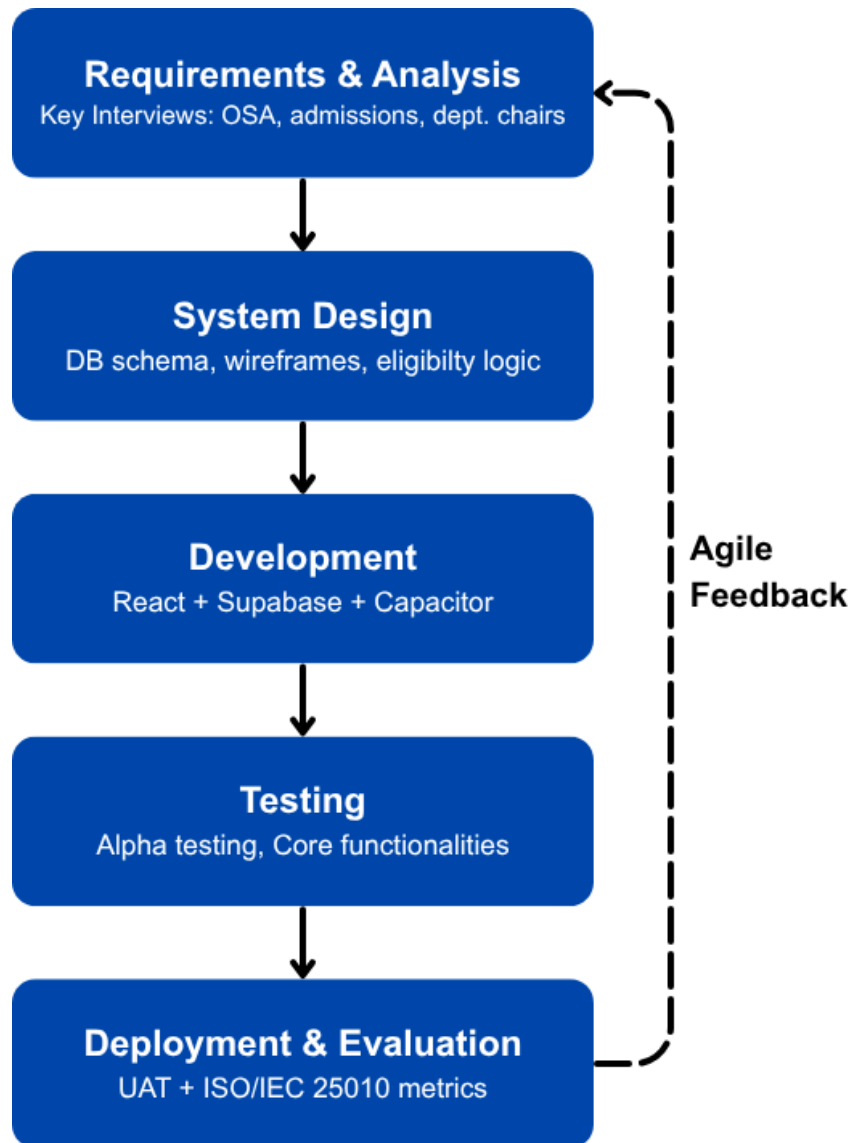


Figure 2: *ScholarPath AdDU Agile SDLC Flowchart*

Phase 1: Requirements Gathering and Analysis. The first phase involves initial data collection through Key Informant Interviews (KII) conducted with primary stakeholders, including OSA administrators, Admissions staff, and Department Chairs. This phase identifies the system's exact functional requirements, existing manual bottlenecks (e.g., Form 230-SCH handling), and the specific student data attributes necessary for the automated scholarship matching algorithm.

Phase 2: System Design. Drawing from the gathered requirements, this phase produces the architectural blueprint of the system. Deliverables include the relational database schema, User Interface/User Experience (UI/UX) wireframes for both student and administrator views, and the logic flow underpinning the Smart Eligibility Checker.

Phase 3: Development. The core programming phase translates design artifacts into a functioning system. Front-end interfaces are built using Vite as the primary build tool, paired with a component-based JavaScript framework (React or Vue) for a reactive, modular UI. Back-end infrastructure — including the PostgreSQL database, secure Document Vault storage, and user authentication — is managed entirely through Supabase. To extend the application to mobile platforms, Capacitor wraps the Vite production build into native Android and iOS packages, enabling cross-platform deployment without maintaining a separate codebase.

Phase 4: Testing. Prior to wider release, the system undergoes alpha testing to verify that core functionalities — such as document uploading and automated criteria matching — operate without critical errors. Issues surfaced during this phase are documented and fed back into development for resolution before proceeding.

Phase 5: Deployment and Evaluation. The functional prototype is deployed to a purposive sample of end-users for User Acceptance Testing (UAT). System quality is then assessed using standardized usability metrics defined by the ISO/IEC 25010 standard, providing quantitative and qualitative data to inform further refinement.

As the feedback loop in Figure 2 indicates, findings from the evaluation phase are not terminal — they inform subsequent iterations, allowing requirements, design decisions, and development priorities to be revisited as stakeholder needs become clearer through real-world use.

3.3 System Architecture and Algorithmic Logic

The architectural decisions underlying ScholarPath AdDU are not derived independently — each major component is a direct response to a documented gap in the reviewed literature. The choice of a centralized Supabase/PostgreSQL backend was motivated by the policy desynchronization documented in Catibog [16] and Orgianus et al. [36], where decentralized, multi-campus systems produced deadline mismatches and contradictory eligibility information that a single canonical database is architecturally positioned to eliminate. The Document Vault — a one-time upload, metadata-tagged, multi-attach storage architecture — directly addresses the integrated document handling gap identified independently by both Al-Ayyubi and Maulana [3] and Bagaporo and Espinosa [12], whose systems demonstrated measurable processing improvements while still requiring physical document submission as a separate, offline step. The Rule-Based Eligibility Engine with its Exclusion Flag Hierarchy was motivated by Bilog's [14] finding that student self-assessment of eligibility was the primary driver of application errors in multi-campus deployments, and that sequential rule evaluation without conflict resolution produced edge cases where students with concurrent government grants passed eligibility checks for additional state-sponsored awards. The Dynamic Faceted Search with compound B-Tree and Range indexing was grounded in Mahdi et al.'s [4] confirmation that faceted mechanisms outperform keyword retrieval in repositories with multiple intersecting categorical attributes, and Tunkelang's [43] finding that the quality of the underlying metadata taxonomy determines whether faceted search reduces or merely redistributes cognitive load. The automated multi-channel notification subsystem responds to the missed-deadline patterns attributable to passive communication channels, documented as a structural weakness in decentralized administration by Catibog [16]. Finally, the hybrid

mobile-and-web architecture via the Capacitor wrapper was directly motivated by the World Bank's [48] caution that digitization without mobile-first design fails to achieve adoption in Philippine HEI contexts, particularly for Mindanao-based student populations. These connections are what distinguish ScholarPath AdDU's design as an evidence-grounded architectural response rather than a general-purpose digitization exercise.

3.3.1 The "Smart Eligibility Checker" (Automated Matching Engine)

This engine actively evaluates three primary student parameters — Quality Point Index (QPI), household income, and enrolled degree program — against the specific constraints of each grant. Student QPI is self-reported during profile registration (see Task 1, Section 3.4.1) and stored in the system's PostgreSQL database. The platform does not integrate with the AdDU University Registrar's database; QPI data accuracy is therefore dependent on the student's own input at the time of registration and is not programmatically verified against official academic records prior to eligibility matching.

To handle institutional edge cases and policy contradictions—such as desynchronized guidelines where a general STEM grant might implicitly exclude BS Nursing due to separate college undergraduate -specific clinical fees—the engine utilizes an **Exclusion Flag Hierarchy**. In this logic, specific negative constraints (exclusions) always override general positive constraints (inclusions).

Pseudocode for the Matching Algorithm:

The following pseudocode illustrates the sequential logic processed by the backend engine when a student requests a scholarship match:

JavaScript

```
FUNCTION ExecuteSmartMatch(StudentProfile, ScholarshipDatabase):
```

```
    Initialize Array EligibleGrants = []
```

```
    // Step 1: Extract Student Attributes
```

```
    SET s_QPI = StudentProfile.QPI
```

```
    SET s_Income = StudentProfile.HouseholdIncome
```

```
    SET s_Degree = StudentProfile.DegreeProgram
```

```
    SET s_HasActiveGovGrant = StudentProfile.HasActiveGovernmentGrant
```

```

// Boolean flag; TRUE if student holds a concurrent state-sponsored award

// Step 2: Iterate through the funding pipelines
FOR EACH Grant IN ScholarshipDatabase:

    // Skip inactive or expired grants

    IF Grant.IsActive == FALSE OR Grant.Deadline < CURRENT_DATE THEN

        CONTINUE

    END IF

    // Step 3: Base Criteria Evaluation

    IF s_QPI >= Grant.MinimumQPI AND s_Income <= Grant.MaximumIncome THEN

        // Step 4: Degree Program Matching

        IF s_Degree IN Grant.EligibleDegrees OR Grant.EligibleDegrees ==
"ALL" THEN

            // Step 5: Policy Contradiction & Exclusion Resolution

            // Example: Resolving concurrent grant restrictions

            IF s_HasActiveGovGrant == TRUE AND Grant.AllowsMultipleGrants
== FALSE THEN

                CONTINUE // Exclusion overrides eligibility; skip grant

            END IF

```

```

        // If all checks pass, append to user's dashboard

        EligibleGrants.APPEND(Grant)

    END IF

END IF

ENDFOR

// Step 6: Compute fit-margin relevance score for secondary sorting

FOR EACH Grant IN EligibleGrants:

    SET margin_QPI    = s_QPI    - Grant.MinimumQPI

    // Higher = stronger academic fit

    SET margin_Income = Grant.MaximumIncome - s_Income

    // Higher = stronger financial fit

    // Normalize and combine margins into a single relevance score.

    // Weights can be adjusted; equal weighting used here as a baseline.

    SET Grant.FitScore = (margin_QPI * 0.5) + (margin_Income / 10000 *
0.5)

ENDFOR

```

```
// Step 7: Return list sorted by deadline (primary), then by fit score
(secondary)

// Primary sort ensures time-critical grants appear first;

// secondary sort differentiates between equally urgent grants by match
strength.

RETURN SORT(EligibleGrants,

            PRIMARY BY Grant.Deadline ASCENDING,

            SECONDARY BY Grant.FitScore DESCENDING)

ENDFUNCTION
```

To provide a clearer visualization of this process, the sequential logic and decision gates of the automated matching engine are mapped out in Figure 3 below.

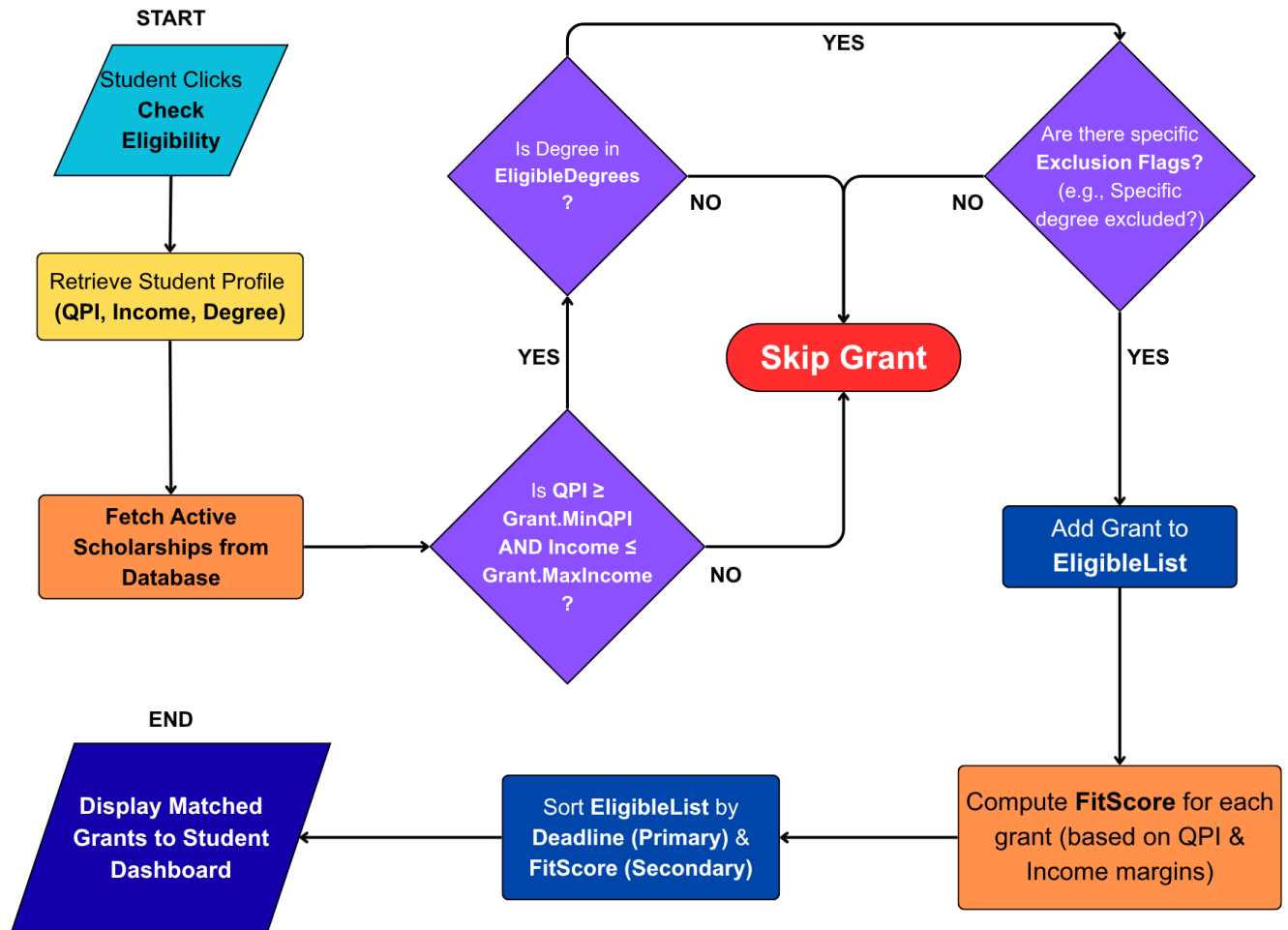


Figure 3: Logic Flowchart of the ScholarPath AdDU Smart Eligibility Checker.

As illustrated in Figure 3, the algorithm prioritizes the evaluation of hard constraints (QPI and household income) before moving to conditional constraints (degree program and exclusion flags). This structured flow ensures that the system efficiently filters out incompatible grants early in the process, minimizing database load and preventing mismatched applications from reaching the student's dashboard. Grants that pass all constraint checks are not returned on equal footing: the algorithm computes a fit-margin relevance score for each eligible grant, calculated from the degree to which the student's QPI exceeds the grant's minimum threshold and the degree to which the student's household income falls below the grant's maximum ceiling. The final output is sorted by deadline (ascending) as the primary key — ensuring time-critical opportunities appear first — and by fit score (descending) as the secondary key, so that among equally urgent grants, those where the student's profile most strongly satisfies the criteria are ranked higher. This differentiates ScholarPath AdDU from a binary pass/fail filter and provides students with a relevance-ordered, not merely deadline-ordered, scholarship dashboard.

3.3.2 Dynamic Faceted Search and Backend Indexing Strategy

To resolve the discovery friction caused by relying on basic keyword searches, ScholarPath AdDU employs a Dynamic Faceted Search architecture. This allows students to apply multiple intersecting

filters (e.g., "State-Sponsored Grants" + "Active Deadlines" + "Full Tuition Coverage") to instantly narrow down the 54 available pipelines without system latency.

Backend Database Categorization: The database normalizes the scholarship data by strictly categorizing the 54 pipelines into four primary metadata facets based on their funding origin: Internally Funded Endowments, Corporate & External Foundations, State-Sponsored Grants, and Specialized Service Pipelines. (The exhaustive tabular dataset utilized to construct these database facets is detailed in Appendix A).

Indexing Strategy for Real-Time Retrieval: To ensure real-time rendering without lag, the system's backend database (e.g., Supabase) utilizes **Compound Indexing**. Instead of scanning the entire database for every search query time complexity), the database pre-sorts records based on frequently queried combinations.

- **B-Tree Indexes** are applied to exact-match categorical fields (e.g., Grant_Category and Status).
- **Range Indexes** are applied to continuous numerical fields (e.g., QPI_Requirement and Application_Deadline).
- **Client-Side Caching & Debouncing:** The frontend interface utilizes asynchronous JavaScript (AJAX/Fetch) combined with an input debounce of 300 milliseconds. This ensures that the database is only queried after the user finishes typing or selecting facets, drastically reducing server load and preventing UI stuttering.

3.3.3 System Architecture and Cloud Infrastructure

To resolve the operational bottlenecks caused by the physical submission of dense paperwork, such as Form 230-SCH and government-stamped tax returns, ScholarPath AdDU employs a secure, cloud-based architecture. The integration of the database, storage, and external communication services is mapped in Figure 4 below.

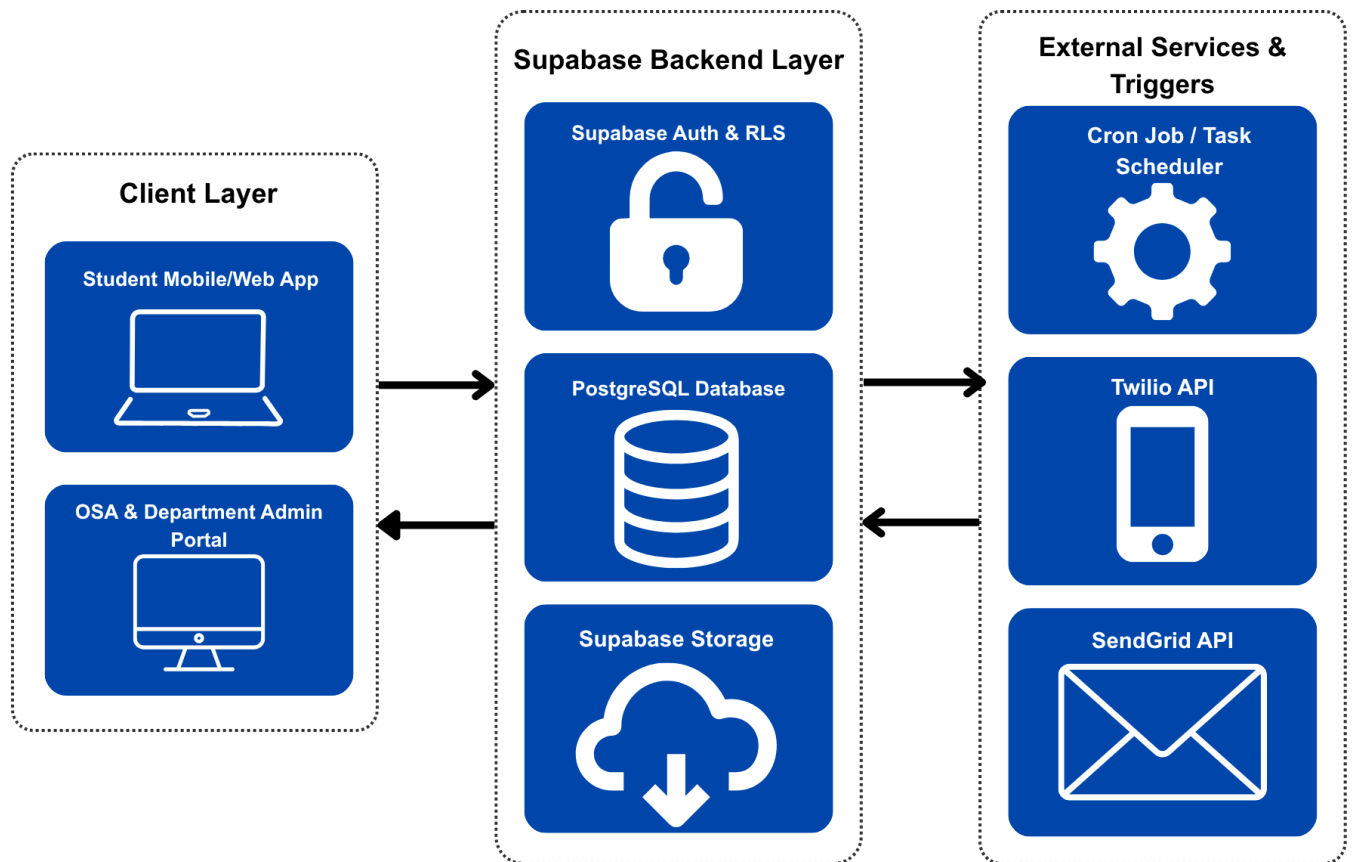


Figure 4: *High-Level System Architecture and Data Flow of ScholarPath AdDU.*

Technical Stack and Storage: The system is built on a three-layer technical stack selected for its cohesion and rapid prototyping capabilities. At the client layer, Vite serves as the frontend build tool, paired with a component-based JavaScript framework (React or Vue) to produce a fast, modular web interface. For cross-platform mobile deployment, Capacitor wraps this same Vite production build into native Android and iOS applications, eliminating the need to maintain a separate mobile codebase. At the backend layer, Supabase serves as the core Backend-as-a-Service (BaaS) infrastructure, providing a hosted PostgreSQL relational database for system state and metadata, Supabase Storage for large object files (i.e., the Document Vault), and Supabase Auth integrated natively with Row Level Security (RLS) policies. This unified ecosystem was selected for its seamless integration of database, storage, and security layers within a single managed service.

Metadata Tagging and Virtual Attachment: To eliminate redundant file uploads, the Document Vault operates on a "one-time upload, multi-attach" architecture. When a student uploads a prerequisite document, the physical file is securely stored exactly once in a dedicated Supabase Storage bucket. The system generates a unique, secure file path for this object.

Instead of duplicating the physical file for every new scholarship application, the PostgreSQL database utilizes metadata tagging. A Documents table records the file's metadata (e.g., StudentID, DocumentType, UploadTimestamp, and the Storage_Path). When a student applies for multiple

scholarships, the system simply creates a relational link (a foreign key) in the Applications table pointing back to the single original document record. This highly normalized architecture ensures optimized storage limits and rapid retrieval times for the Office of Student Affairs (OSA).

Income Document Verification Workflow: The Document Vault's handling of proof-of-income documents follows a structured, administrator-validated verification process to ensure that scholarship applications are not processed on the basis of unreviewed financial data. When a student uploads a proof-of-income document — such as a BIR-stamped Income Tax Return (ITR) — to the Document Vault, the document is automatically assigned a verification status of *Pending* in the Documents table. An OSA Administrator subsequently reviews the uploaded document through the admin dashboard, examining it against the requirements of the applicable scholarship. The administrator then manually updates the document's verification status to either *Verified* or *Rejected* based on their assessment. Scholarship applications linked to documents with a verification status of *Pending* or *Rejected* cannot proceed to final processing by the OSA. This workflow ensures that the eligibility matching engine operates on validated financial data while preserving the administrative authority of the OSA over the authenticity determination — a distinction that cannot be automated without integration with official government tax databases, which is outside the current system scope.

3.3.4 Role-Based Access Control (RBAC) and Data Privacy Compliance

Because the ScholarPath AdDU platform processes highly sensitive economic and academic data, its architecture is fundamentally governed by the Philippine Data Privacy Act of 2012 (RA 10173).

Encryption and Row Level Security (RLS): To ensure strict compliance and secure data handling, the system relies on Supabase Auth and PostgreSQL's native security features. All data in transit is secured using Transport Layer Security (TLS). Furthermore, access to sensitive data at rest—both within the database tables and the Supabase Storage buckets—is restricted using Row Level Security (RLS) policies. RLS ensures that access controls are enforced directly at the database level, preventing unauthorized queries regardless of how the frontend application is accessed.

RBAC Permissions Matrix: To enforce the principle of least privilege, the system utilizes a strict Role-Based Access Control (RBAC) architecture. Upon authentication, users are assigned distinct, immutable privileges based on their specific role within the university, which are directly mapped to the database RLS policies.

Table 3: ScholarPath AdDU Role-Based Access Control (RBAC) Matrix

User Role	Data Access Privileges	Core System Actions
-----------	------------------------	---------------------

Undergraduate Students	Read/Write (Self Only): Can view and edit their own academic/financial profiles and uploaded documents. Blocked by RLS from viewing other users' data.	Search for grants, upload files to personal Document Vault, submit applications, view application status.
OSA Administrators	Read/Write (Global): Full authorized access to view all submitted student applications, verified documents, and system-wide scholarship parameters.	Approve/reject applications; review uploaded proof-of-income documents and manually update verification status to <i>Verified</i> or <i>Rejected</i> via the admin dashboard (documents remain in <i>Pending</i> status until this action is taken, blocking final application processing); broadcast announcements; trigger manual notifications; extract analytical reports.
Department Chairs & Coordinators	Read (Restricted): Can only view the aggregated academic and economic profiles of applicants within their specific department.	Filter and assess economic statuses specifically for Grant-in-Aid (GIA) selections, submit official endorsements to OSA.

3.3.5 Automated Notification Subsystem

The absence of proactive alerts frequently results in missed application deadlines and subsequent student attrition. To rectify this, ScholarPath AdDU integrates an automated, multi-channel notification subsystem designed to deliver critical deadline alerts and status updates via SMS and email, without requiring manual administrative intervention.

Third-Party API Integration: The system leverages enterprise-grade communication APIs to ensure reliable message delivery. The **SendGrid API** is utilized to format and dispatch detailed, formal email notifications regarding application outcomes and institutional updates. For time-sensitive alerts, such as approaching 48-hour deadlines, the system integrates the **Twilio SMS API** to push brief, immediate text messages directly to the student's registered mobile device.

Automated Trigger Logic and Edge Functions: To interface with the Supabase backend, the notification engine is driven by automated task schedulers (Cron jobs) and Supabase Edge Functions.

The trigger logic evaluates two primary conditions:

1. **Deadline Proximity:** A scheduled backend worker queries the database daily. If the Current_Date is within a predefined threshold (e.g., 3 days) of a Grant.Deadline AND the user has an application marked as Status = 'Draft', the system executes an Edge Function to send a "Deadline Reminder" payload via the Twilio API.
2. **Status Mutation:** Whenever an OSA Administrator changes an application status in the database (e.g., from 'Pending' to 'Approved'), a database webhook immediately triggers an Edge Function. This function compiles the relevant context and pushes a "Status Update" payload to the SendGrid API, delivering the official results to the student's inbox.

3.4 Testing & Evaluation Procedures: ISO/IEC 25010 Task-Based Usability Protocol

To assess the functional suitability and usability of the ScholarPath AdDU system, the study employs a mixed-method evaluation approach. This methodology combines task-based observational testing with a standardized System Usability Scale (SUS) questionnaire, administered to a purposive sample of approximately 10 to 15 end-users. This sample size ensures sufficient data saturation and will consist of undergraduate students, active Office of Student Affairs (OSA) administrators, and selected Department Chairs to guarantee comprehensive coverage of the system's multi-role interfaces.

3.4.1 Step-by-Step Testing Procedure

The evaluation will be conducted in a controlled environment following a standardized script to prevent researcher bias. The procedure is outlined as follows:

1. **Briefing and Consent:** Participants are introduced to the system's purpose. The facilitator explicitly states that the system is being tested, not the user, and secures informed consent.
2. **Task Execution:** Participants are asked to complete four specific, real-world scenarios designed to test core system features. The facilitator observes silently without offering assistance unless a critical failure occurs.
 - **Task 1: Profile Registration:** The user must create a new account, inputting required academic and socioeconomic data, such as QPI, household income, and degree program.
 - **Task 2: Document Upload:** The user navigates to the Document Vault and successfully uploads a standardized test file (e.g., a mock Form 230-SCH) , verifying that the system securely stores the file and registers the metadata appropriately.
 - **Task 3: Dynamic Search and Application:** The user utilizes the dynamic faceted search to filter the available funding pipelines to locate a specific "Corporate & External Foundation" grant and successfully submit an application.
 - **Task 4: Admin Dashboard Navigation (OSA Admin Only):** The user logs into the admin portal , navigates to pending applications, and changes the status of a specific student application from "Pending" to "Approved" to trigger the automated notification subsystem.
3. **Observational Data Collection:** During the execution of Tasks 1 through 4, the facilitator records specific usability metrics (detailed in Section 3.4.2).

4. **Post-Session Instruments:** Immediately after completing the four tasks, participants are administered two sequential instruments in the following order. First, the participant completes the standardized **10-item System Usability Scale (SUS)** questionnaire to assess overall subjective perceived usability of the ScholarPath AdDU interface. The SUS is administered first because it captures the participant's holistic impression of the system immediately after task completion, before any comparative framing is introduced. Second, the participant completes the **Post-Prototype Perceived Effort Survey (Appendix C)**, which asks them to estimate the time and effort required by the digital process and to compare that perception against the manual experience they described in the pre-study. These are presented as two distinct instruments and are not to be conflated: the SUS measures general interface usability, while Appendix C generates the comparative data used to answer Research Questions 1, 2, and 3.
5. **Debriefing:** A brief, semi-structured exit interview is conducted to gather qualitative feedback on any frustrations or difficulties encountered.

3.4.2 Data Analysis and Statistical Tools.

To objectively quantify the system's performance and user satisfaction, the following metrics and statistical tools will be utilized:

- **Quantitative Observational Metrics:**
 - **Time-on-Task:** Measured in seconds, this records the exact duration a user takes to successfully complete each of the four designated tasks.
 - **Error Rates:** Calculated as the number of misclicks, failed uploads, or incorrect navigation paths taken per task, as well as the overall percentage of users who fail to complete a task independently.
 - **Notification Delivery Success Rate:** Measured during Task 4 (Admin Dashboard Navigation, Section 3.4.1), this metric records the percentage of triggered notification events — specifically, application status mutations from 'Pending' to 'Approved' executed by the OSA Administrator participant — that result in a successfully dispatched notification to the student's registered contact channel (SMS via Twilio or email via SendGrid) within the testing session. The facilitator confirms delivery by verifying receipt on the designated student-side test device. The metric is computed as: $(\text{Number of Successfully Delivered Notifications} / \text{Total Triggered Notification Events}) \times 100$. A delivery success rate of 100% across the testing session is the target threshold, given the controlled, low-concurrency testing environment.
 - **Trigger-to-Delivery Latency:** Recorded in seconds, this measures the elapsed time between the moment an OSA Administrator participant commits a status mutation in the admin dashboard (the event trigger) and the moment the corresponding notification is confirmed as received on the student-side test device (the delivery confirmation). The facilitator records both timestamps manually during Task 4. Mean latency across all triggered events in the session is computed and reported. This metric operationalizes the performance efficiency dimension of the notification subsystem as distinct from overall system response time.
- **System Usability Scale (SUS) Analysis:**

- The SUS provides a validated instrument for measuring participants' subjective perceived usability of the platform. Responses are gathered on a standard 5-point Likert scale across 10 items, with alternating positive and negative phrasing to reduce response bias. Raw SUS scores, ranging from 0 to 100, will be analyzed using descriptive statistics. The **Mean** will be calculated to determine the average perceived usability score across the purposive sample, providing an overall benchmark of system acceptability. To contextualize the resulting mean SUS score, the study will apply the adjective rating scale validated by Bangor et al. [13], which maps numerical ranges to descriptive quality grades: scores of 85 and above are classified as Excellent, 71–84 as Good, 50–70 as OK, and below 50 as indicative of usability failure requiring significant redesign. A target mean SUS score of 71 or higher — the Good threshold — is set as the minimum benchmark for the ScholarPath AdDU prototype to be considered functionally acceptable for its intended user base. The **Standard Deviation** will measure the variance in scores, indicating the degree of consensus among participants regarding the system's subjective usability. It is important to note that the SUS addresses Research Question 4 specifically — overall system usability and performance efficiency — and is analytically distinct from the pre/post comparative instruments described below, which address Research Questions 1 through 3.
- **Functional Correctness Evaluation (Smart Eligibility Checker Test Case Matrix)**
 - To directly assess whether the automated matching engine executes the Exclusion Flag Hierarchy without generating false positives or incorrect recommendations — corresponding to the Functional Correctness sub-characteristic in Table 8 (ISO/IEC 25010) — the study employs a structured test case matrix administered separately from the user-facing usability session. Fifteen representative synthetic student profiles are constructed to cover the full range of eligibility edge cases: (1) QPI exactly at the grant's minimum threshold; (2) QPI one decimal point below the minimum threshold; (3) household income exactly at the grant's maximum ceiling; (4) household income one unit above the ceiling; (5) degree program explicitly listed in a grant's eligible degrees; (6) degree program explicitly listed in a grant's exclusion list; (7) degree program absent from both eligible and excluded lists where the grant specifies "ALL"; (8) student holding a concurrent active government grant applying to a grant where `AllowsMultipleGrants = FALSE`; (9) student holding a concurrent active government grant applying to a grant where `AllowsMultipleGrants = TRUE`; (10) student profile satisfying all base criteria but triggering a degree-specific exclusion flag; (11) an expired grant (`Deadline < CURRENT_DATE`) that should be skipped regardless of eligibility; (12) a grant marked `IsActive = FALSE`; (13) a student meeting no criteria for any grant (null result expected); (14) a student meeting criteria for multiple grants simultaneously, verifying correct multi-result output; and (15) a student whose profile satisfies both a broadly inclusive rule and a specific exclusion flag simultaneously, verifying that the exclusion unconditionally overrides the inclusion.

- For each profile, the expected system output — either a specific set of matched grants or an empty result — is documented prior to test execution. The actual system output is then recorded and compared against the expected output. A match constitutes a pass; any discrepancy constitutes a failure and is logged for root cause analysis. The Functional Correctness score is computed as: $(\text{Number of Passing Test Cases} / \text{Total Test Cases}) \times 100$. A minimum pass rate of 100% on the exclusion-flag test cases (profiles 6, 8, 10, 11, 12) is set as the non-negotiable threshold, as false positives on exclusion logic constitute the highest-risk failure mode of the matching engine.
- **Pre/Post Comparative Survey Analysis:**
 - The Pre-Study Student Problem Validation Survey (Appendix B) and the Post-Prototype Perceived Effort Survey (Appendix C) together constitute the comparative data collection instruments for Research Questions 1, 2, and 3. Because both instruments use structurally parallel items with identical scale anchors, their data is analyzed through direct item-level mean comparisons using descriptive statistics.
 - For Component 1 of both instruments (the ordinal time-estimation scale), the distribution of responses across the five ordinal ranges will be tabulated for both the pre- and post-prototype conditions. The modal response category and the direction of shift — whether participants perceive the digital process as taking less, equal, or more time — will be reported to address RQ1.
 - For Component 2 of both instruments (the six-item perceived effort Likert grid), the Mean score per sub-task item will be calculated for both conditions and presented in a side-by-side comparison table. A decrease in mean score from pre- to post-prototype on a given sub-task indicates a perceived reduction in effort for that application stage. Sub-task items 1 and 3 through 5 inform RQ1; sub-task item 2 (eligibility criteria determination) informs RQ2; and sub-task item 6 (deadline tracking) informs RQ3.
 - For Component 3 of the pre-study instrument and Component 3 of the post-prototype instrument, frequency distributions will be used for the categorical items concerning application mismatch history (RQ2) and deadline-miss history (RQ3). The proportion of participants who reported previously submitting ineligible applications will be compared against the proportion who indicated the Smart Eligibility Checker would prevent such errors. Similarly, the proportion who reported previously missing deadlines will be compared against post-prototype responses on the notification subsystem's perceived effectiveness.
 - For Component 4 of the post-prototype instrument (the overall comparative perception item), a frequency distribution of responses across the five-point scale — from Much Less Effort to Much More Effort — will be reported as a summary indicator of the system's perceived overall impact, synthesizing the findings of RQ1 through RQ3.
- **Qualitative Data Analysis (Thematic Analysis):** While the SUS provides a quantitative benchmark, the qualitative feedback gathered during the post-task semi-structured exit interviews will be processed using thematic analysis. The researchers will transcribe user

comments and categorize them into recurring themes (e.g., "Navigation Friction," "Terminology Confusion," or "Positive Interface Feedback"). Identifying these qualitative themes will provide contextual explanations for the quantitative error rates and guide the final iterative refinements of the system's user interface prior to deployment.

3.4.3 Pre-Study Student Problem Validation Survey and Post-Prototype Perceived Effort Survey

To operationalize the within-subjects comparative design described in Section 3.1, this study employs two structurally paired survey instruments administered at distinct phases of the research process. The Pre-Study Student Problem Validation Survey (Appendix B) is administered prior to system exposure to establish baseline data on the perceived burden of the existing manual scholarship application process. The Post-Prototype Perceived Effort Survey (Appendix C) is administered immediately following the task-based usability testing session to capture participants' post-exposure perceptions using an identical scale structure. Together, these instruments generate the comparative data required to address Research Questions 1, 2, and 3, as the System Usability Scale administered in the same session addresses only RQ4.

The paired design deliberately uses within-subjects administration — the same participants complete both instruments — to eliminate inter-group variability. Because each participant's pre-study responses serve as their own individual baseline, any observed reduction in perceived effort or time burden can be attributed to the system's features rather than differences in prior scholarship experience across participants. Both instruments use scaled and categorical response formats rather than open recall of specific durations, acknowledging that the manual application process unfolds over days or weeks and is not conducive to precise time measurement.

Pre-Study Student Problem Validation Survey (Appendix B)

The pre-study instrument is administered to a purposive sample of current or former AdDU scholarship applicants prior to prototype development or any system exposure. It is organized into three components, each targeting a distinct dimension of the manual process and a corresponding research question.

Component 1: Perceived Time-on-Process asks respondents to estimate the total duration of their end-to-end manual application experience — from initial scholarship discovery through physical document submission — using five ordinal ranges: less than 1 day, 1–3 days, 4–7 days, 1–2 weeks, and more than 2 weeks. This item establishes the time-burden baseline for RQ1. It should be noted, however, that post-prototype time estimates are derived from a controlled prototype testing session and may not accurately reflect the duration of the process in actual real-world deployment, where variables such as document readiness, network conditions, and platform familiarity over sustained use may produce different results.

Component 2: Perceived Effort Scale applies a five-point Likert format (1 = Very Easy, 5 = Very Difficult) to six discrete application sub-tasks. The six sub-tasks are: (1) finding out which specific scholarships the student was eligible for; (2) determining whether they specifically met all eligibility

criteria — including QPI thresholds, household income ceilings, and degree program restrictions — for a given scholarship prior to submitting; (3) gathering, printing, and preparing the required physical documents; (4) physically submitting the completed application package to the respective offices; (5) tracking the status of the application after submission; and (6) keeping track of multiple scholarship deadlines simultaneously throughout the application period. Sub-tasks 1, 3, 4, and 5 together establish the effort baseline for RQ1. Sub-task 2 specifically establishes the eligibility confusion baseline for RQ2. Sub-task 6 establishes the deadline management difficulty baseline for RQ3.

Component 3: Application Accuracy and Deadline History uses categorical and multiple-choice items to establish behavioral frequency baselines for RQ2 and RQ3. Respondents are asked whether they have ever submitted an application and later discovered they did not meet the eligibility requirements for that scholarship, and if so, how many times this occurred. They are separately asked whether they have ever missed a scholarship application deadline under the manual process, and if so, what the primary reason was. These items produce the concrete behavioral baseline data against which the post-prototype instrument's feature-effectiveness responses are compared.

Post-Prototype Perceived Effort Survey (Appendix C)

The post-prototype instrument is administered to the same participants immediately following the completion of the task-based usability testing session and after the SUS questionnaire. Its structural mirroring of Appendix B is deliberate and essential: because both instruments use identical Likert anchors, ordinal ranges, and sub-task categories, their corresponding items can be directly compared without transformation.

Component 1 presents the same five ordinal time-estimation ranges as the pre-study, asking participants to estimate how much total time the digital scholarship application process through ScholarPath AdDU would require. This mirrors the pre-study Component 1 for RQ1 comparison.

Component 2 presents the same six perceived effort sub-tasks as the pre-study, reworded to reflect ScholarPath AdDU's specific features: scholarship discovery through the centralized repository; eligibility determination through the Smart Eligibility Checker; document management through the Document Vault; digital application submission; application status tracking through the student dashboard; and deadline awareness through the automated notification subsystem. The same five-point Likert anchors are used, allowing mean scores per sub-task to be compared directly against pre-study means.

Component 3: System Feature Effectiveness moves beyond effort estimation to assess whether participants believe the system's specific features address the problems identified in the pre-study's Component 3. Participants indicate whether they believe the Smart Eligibility Checker would prevent them from applying to scholarships for which they do not qualify — directly addressing the mismatch baseline from RQ2 — and rate the extent to which the automated notification system would help them remain aware of upcoming deadlines — directly addressing the missed-deadline baseline from RQ3.

Component 4: Overall Comparative Perception asks participants to characterize the total effort required by ScholarPath AdDU relative to the traditional manual process, using a five-point scale ranging from Much Less Effort to Much More Effort. This item synthesizes the findings of RQ1 through RQ3 into a single directional indicator and provides a holistic perception anchor for the study's comparative conclusions. An optional open-ended item closes the instrument, inviting participants to identify which specific system feature most reduced the difficulty of the scholarship application process for them.

3.5 Ethical Considerations

Because the usability testing of ScholarPath AdDU involves human participants and the simulated input of sensitive socioeconomic data, the evaluation phase is strictly governed by established research ethics and the Philippine Data Privacy Act of 2012 (RA 10173). The following ethical protocols will be implemented throughout the testing phase:

- **Informed Consent and Voluntary Participation:** Prior to any system interaction, all participants will be provided with an informed consent form detailing the purpose of the study, the specific tasks required, and the estimated time commitment. Participation is strictly voluntary, and end-users maintain the right to withdraw from the testing session at any point without penalty or need for justification.
- **Data Anonymization and Privacy:** During the task-based execution (specifically Task 1: Profile Registration), participants will be explicitly instructed to use provided mock data or pseudonymized profiles rather than their actual household income or academic records. This ensures that no genuine sensitive information is exposed or recorded during the evaluation of the system's matching algorithm.
- **Confidentiality and Secure Data Disposal:** All observational metrics; pre-study and post-prototype survey responses (Appendices B and C), and SUS questionnaire responses will be kept strictly confidential. The qualitative and quantitative data will be aggregated so that no single metric or quote can be traced back to an individual participant. Upon the successful defense and final submission of the capstone project, all test accounts, mock databases, and physical consent forms utilized during the evaluation phase will be permanently and securely deleted to prevent any unauthorized data retention.

Chapter 4: THEORETICAL BACKGROUND

4.1 Introduction

The design and implementation of ScholarPath AdDU is grounded in nine established theoretical frameworks drawn from computer science, information science, and software engineering. Together, these theories provide formal justification for every major architectural decision — from database structure and document management to security, user experience, and development methodology.

This chapter situates each theory within the overarching Information Systems Success Model [20] adopted in Chapter 2. Relational Database Theory and Metadata-Driven Document Management underpin System Quality by ensuring structural integrity and storage efficiency. Information Foraging

Theory and Event-Driven Architecture support Information and Service Quality through improved discoverability and proactive communication. Rule-Based Expert Systems Theory operationalizes the automated eligibility matching engine. Finally, Cloud Infrastructure Theory, RBAC, Empirical Process Control, and the HCI/ISO 25010 framework collectively ensure scalability, legal compliance, iterative development, and empirically validated usability.

4.2 Relational Database Theory and Normalization

Consolidating student academic profiles, household financial metrics, and the eligibility criteria for 54 scholarship pipelines into a consistent repository requires a rigorous structural foundation. Relational Database Theory — specifically the normalization principles formalized by Codd [17] — provides this foundation by mathematically organizing data into interconnected two-dimensional tables governed by first-order predicate logic. Within the DeLone and McLean [20] framework, a normalized 3NF schema is the foundational enabler of System Quality — specifically the sub-dimension of data integrity — as it eliminates the inconsistencies that would otherwise corrupt the automated matching engine's outputs.

4.2.1 The Relational Model and Anomaly Prevention

Codd [17] defined normalization as the systematic process of organizing relations to reduce redundancy and prevent modification anomalies. In the AdDU context, three anomaly types directly threaten the integrity of an unnormalized financial aid database. Table 4.1 outlines each anomaly and its resolution within ScholarPath AdDU.

Table 4: Database Anomalies and Their Resolution in ScholarPath AdDU

Anomaly Type	Definition	Resolution in ScholarPath AdDU
InsertionAnomaly	Facts cannot be recorded because they lack a required primary key from another entity.	Scholarship providers are stored independently of student application records. New grants (e.g., a new SM Foundation program) are added to the database before any student applies.
UpdateAnomaly	A single logical fact recorded across multiple rows creates contradictions when updates are partially executed.	Each scholarship deadline and criterion exists in one canonical row. Any update propagates instantly across all 54 pipelines and concurrent user queries.
DeletionAnomaly	Deleting one record unintentionally destroys data representing an entirely different, independent fact.	Archiving a graduated student's profile does not affect scholarship criteria, which are stored in fully independent relations.

4.2.2 Progression Through Normal Forms

The ScholarPath AdDU schema adheres to the Third Normal Form (3NF), progressing through Codd's sequential requirements. First Normal Form (1NF) mandates atomic, scalar values with no repeating groups; prerequisite documents such as Form 230-SCH and Income Tax Returns are therefore stored as discrete, queryable records rather than comma-separated strings. Second Normal Form (2NF) eliminates partial dependencies, ensuring grant provider details are stored independently of student application records. Third Normal Form (3NF) eliminates transitive dependencies, ensuring — as Kent [32] summarizes — that every non-key attribute provides "a fact about the key, the whole key, and nothing but the key." Adhering to 3NF guarantees that any update to a scholarship attribute, such as a revised QPI threshold or an extended deadline, propagates instantly from a single authoritative record to all related queries.

4.3 Centralized Information Systems and Cloud Infrastructure Theory

The decision to deploy ScholarPath AdDU as a single unified web platform — rather than distributing data across departmental servers — is grounded in centralized IT infrastructure theory [34]. Centralized infrastructure consolidates computing resources and data into one managed environment, enabling uniform policy enforcement and the elimination of data silos that cause policy desynchronization across departments [41]. By hosting all 54 financial aid pipelines on a single Supabase backend, deadline updates and eligibility changes are immediately reflected for all users simultaneously.

Modern cloud platforms mitigate the traditional risk of centralization — single points of failure — through hardware virtualization and physical redundancy. Mell and Grance [34] define cloud computing as on-demand access to a shared pool of configurable computing resources characterized by rapid elasticity and measured service. For ScholarPath AdDU, Supabase's managed PostgreSQL infrastructure provides high availability alongside immutable audit logs that support compliance with the Philippine Data Privacy Act of 2012 (RA 10173), transforming regulatory requirements into an operational advantage.

In the IS Success Model, centralized cloud infrastructure directly supports System Quality through 24/7 availability and Service Quality through uniform, institution-wide policy enforcement.

4.4 Metadata-Driven Document Management

Manual, paper-based document workflows create severe operational bottlenecks at the AdDU Office of Student Affairs. ScholarPath AdDU resolves this through metadata-driven document management, which decouples physical file storage from logical document associations [25]. Table 5 contrasts traditional hierarchical storage with the metadata-driven approach.

Table 5: Traditional Hierarchical Storage vs. Metadata-Driven Architecture

Feature	Traditional Hierarchical Storage	Metadata-Driven Architecture
Location Strategy	Files are locked into a rigid folder path.	Files are stored in a flat virtual pool; organization is determined dynamically by relational metadata tags.

Feature	Traditional Hierarchical Storage	Metadata-Driven Architecture
Data Duplication	One document submitted to three grants requires copying the physical file three times.	A single file is uploaded once. Multiple scholarship applications reference the same URI via foreign keys.
Retrieval Mechanism	Users must navigate a rigid folder hierarchy to locate files.	Administrators execute high-speed relational queries by document type, upload date, or verification status.

ScholarPath AdDU operationalizes this theory as the Document Vault. When a student uploads a prerequisite file — such as a BIR-stamped Income Tax Return or an academic transcript — it is stored exactly once in a Supabase Storage bucket and assigned a unique URI. Subsequent scholarship applications attach to that original file via foreign keys in the Applications table, eliminating physical duplication and enabling OSA administrators to query and verify documents through metadata tags rather than folder hierarchies.

4.5 Rule-Based Expert Systems Theory

Automating the complex matching of students to eligible scholarships requires codifying the logic of the AdDU College Scholarship Committee as machine-executable rules. Rule-Based Expert Systems (RBS) theory provides the formal mechanism for this, representing human domain expertise as conditional IF-THEN logic [15]. In the DeLone and McLean [20] model, the matching engine operationalizes Information Quality — specifically the accuracy and completeness dimensions — by ensuring that only verifiably eligible grants are surfaced to the student's dashboard.

4.5.1 Knowledge Representation and the Inference Engine

Buchanan and Duda [15] describe expert systems as consisting of three interdependent components: a Knowledge Base of facts and IF-THEN rules, a Working Memory storing the current input state, and an Inference Engine that applies rules to derive conclusions via the logical principle of modus ponens. In ScholarPath AdDU, the Knowledge Base is the normalized database of 54 scholarship pipelines with their eligibility constraints; the Working Memory is the student's profile at the time of a match request; and the Inference Engine applies a forward-chaining (data-driven) strategy, iterating from known student attributes — QPI, household income, and degree program — to an ordered array of eligible grants.

4.5.2 Conflict Resolution and the Exclusion Flag Hierarchy

A core design challenge in inference engines is conflict resolution: determining system behavior when multiple rules simultaneously apply to the same data state. ScholarPath AdDU implements a priority-based strategy called the Exclusion Flag Hierarchy, in which specific negative constraints unconditionally override general positive constraints. For example, a state-sponsored STEM grant may broadly include all students in the Science and Technology cluster, while a separate university policy excludes BS Nursing students due to non-coverable clinical fees. The inference engine

resolves this by applying the exclusion flag and bypassing the grant entirely, regardless of whether the student satisfies QPI and income criteria. Table 6 maps the conflict resolution strategies to their system context.

Table 6: Conflict Resolution Strategies in the Smart Eligibility Checker

Strategy	Theoretical Definition	ScholarPath AdDU Implementation
Complexity Priority	Gives highest priority to rules with the most complex condition clause.	Highly granular financial constraints (income ceilings, dual-grant restrictions) are evaluated before broad academic minimums (degree cluster).
Exclusion Flag Hierarchy	Specific negative constraints unconditionally supersede general positive inclusion rules.	An active exclusion flag — such as a clinical-program exclusion or a dual-active-grant restriction — immediately skips the grant, regardless of passing QPI or income checks.

4.6 Information Foraging Theory and Faceted Search

The cognitive processes through which users navigate complex digital environments are formalized by Information Foraging Theory (IFT), developed by Pirolli and Card [37] at PARC. IFT applies optimal foraging theory from evolutionary ecology to model how users maximize the rate of information gain while minimizing cognitive cost — analogous to how animals optimize food acquisition.

IFT rests on three core constructs: Information Patches (localized clusters of related content within an interface), Information Diet (the specific subset of sources a user pursues to satisfy a goal), and Information Scent (the user's cognitive estimate of how likely a given path is to yield relevant results). When perceived scent drops below a threshold, users abandon the current patch and navigate elsewhere [37].

In the AdDU scholarship context, presenting 54 pipelines as an unstructured list produces weak information scent and high abandonment. ScholarPath AdDU addresses this through Dynamic Faceted Search, which exposes the system's underlying metadata structure directly in the interface. Students intersect facets — such as "State-Sponsored Grants," "Active Deadlines," or "Full Tuition Coverage" — to iteratively refine the information patch and prevent empty result sets [4, 43]. Backend compound indexing and a 300-millisecond client-side debounce ensure instantaneous query response, preserving the low temporal cost critical to effective information foraging.

4.7 Event-Driven Architecture for Asynchronous Notification

The systemic problem of missed deadlines at AdDU is attributable to passive, manual communication channels. ScholarPath AdDU addresses this through Event-Driven Architecture (EDA), which shifts the notification system from a synchronous request-response model to an

asynchronous, event-triggered one [28]. EDA is built around the production, detection, and consumption of events — defined as significant changes in system state — and operates on the principle of loose coupling, where components interact independently with no awareness of each other's internal logic.

By adopting EDA, the notification subsystem operates without blocking the main application server. Table 7 maps the two primary event triggers to their corresponding EDA components.

Table 7: Event-Driven Architecture Trigger Mapping for the Automated Notification Subsystem

EDA Component	Temporal Trigger (Deadline Proximity)	Transactional Trigger (Status Mutation)
Event Producer	Scheduled Cron job querying the database daily for grants within the 72-hour threshold.	OSA administrator modifying an application status in the dashboard.
Event Router	Database trigger logic identifying records nearing the threshold.	Database webhook detecting the state mutation upon commit.
Event Consumer	Serverless Edge Function pushing a time-sensitive alert via the Twilio SMS API.	Edge Function compiling context and executing a status-update payload via the SendGrid Email API.

This decoupled architecture ensures that third-party API latency — inherent in calls to Twilio and SendGrid — is absorbed entirely by the Edge Function layer, allowing the core web server to scale efficiently under concurrent student load.

4.8 Role-Based Access Control and Data Privacy

Financial aid systems process highly sensitive personal and socioeconomic data, necessitating rigorous access control. ScholarPath AdDU implements the NIST Role-Based Access Control (RBAC) model, formalized by Ferraiolo and Kuhn [24] and standardized by NIST in 2004.

RBAC replaced error-prone, per-user Access Control Lists (ACLs) with a tripartite structure: Users are assigned Roles representing their organizational function, and Permissions are attached exclusively to those Roles [38]. This structure inherently enforces two foundational security principles: the Principle of Least Privilege, which limits access to the minimum required for each user's duties, and Separation of Duties, which distributes critical workflows across roles to prevent unilateral fraud or error.

RBAC is the direct theoretical mechanism by which ScholarPath AdDU achieves compliance with RA 10173. Because the Document Vault stores government-stamped ITRs and confidential household income data, a flat access model would constitute a legal liability. Undergraduate students hold self-access only; OSA Administrators hold global administrative access; Department Chairs hold read-only access restricted to their department's applicant pool. These boundaries are enforced at the database engine level through PostgreSQL Row Level Security (RLS) policies,

ensuring that even if an application endpoint is accidentally exposed, the underlying data remains inaccessible without proper role-verified authentication.

Within the IS Success Model, RBAC compliance is the mechanism by which the platform satisfies the Service Quality dimension, specifically the assurance sub-dimension, by guaranteeing that access to sensitive economic data is governed strictly and consistently.

4.9 Empirical Process Control Theory in Agile Development

The development of ScholarPath AdDU is governed by Empirical Process Control theory, the foundational pillar of Agile methodology [39]. Unlike the Waterfall model, which assumes complete upfront specification of all requirements, empirical process control acknowledges that complex software systems operate in volatile, ambiguous environments where requirements emerge through iterative stakeholder interaction.

The theory rests on three interdependent pillars [39]: Transparency, requiring that all development progress and system artifacts be visible and understood by both developers and stakeholders; Inspection, requiring frequent examination of the system against standard metrics to detect variances; and Adaptation, requiring immediate process adjustment when inspection reveals unacceptable deviation.

Because the 54 scholarship pipelines contain undocumented edge cases that cannot be fully specified before development begins, a rigid sequential approach would produce a flawed matching engine. The Agile SDLC described in Chapter 3 operationalizes all three pillars: the functional prototype undergoes alpha testing and User Acceptance Testing (UAT) as the Inspection phase, and feedback from OSA administrators and student testers drives immediate schema adjustments and rule-logic refinements as the Adaptation phase.

In the context of the IS Success Model, empirical process control supports System Quality by enabling continuous verification that the system's functional logic aligns with stakeholder requirements before each phase of testing.

The continued relevance of this empirical approach in contemporary software development projects has been validated in recent literature. Hoda et al. [27] conducted a systematic mapping of agile practices across 1,540 studies, confirming that transparency, iterative inspection, and stakeholder-driven adaptation remain the dominant predictors of project success in environments characterized by evolving or partially specified requirements — conditions that precisely describe the undocumented edge cases within AdDU's 54-pipeline financial aid taxonomy.

4.10 Human-Computer Interaction and the ISO/IEC 25010 Model

While robust backend architecture ensures computational reliability, the practical success of an information system is ultimately determined by user adoption. The interface design and formal evaluation of ScholarPath AdDU are therefore guided by Human-Computer Interaction (HCI) theory and the ISO/IEC 25010:2023 Software Quality Evaluation standard [23, 31].

HCI focuses on designing interactive computing systems that align with users' cognitive capacities and actual workflows [23]. For AdDU students, the legacy process — physically tracking down documents, parsing eligibility criteria across multiple disparate websites, and submitting brown envelopes — imposes unnecessary cognitive overhead. ScholarPath AdDU alleviates this through a centralized dashboard aggregating deadlines and eligibility statuses into a single view, and a

mobile-responsive architecture enabling document uploads from any device, thereby bypassing the physical constraints of OSA office hours.

The ISO/IEC 25010:2023 SQuaRE standard provides the formal evaluation framework, defining product quality across eight measurable characteristics. The ScholarPath AdDU prototype assessment operationalizes two dimensions — Functional Suitability and Performance Efficiency — as detailed in Table 8.

Table 8: ISO/IEC 25010:2023 Evaluation Dimensions for ScholarPath AdDU

ISO/IEC 25010 Characteristic	Sub-Characteristic	Evaluation Metric for ScholarPath AdDU
Functional Suitability	Functional Completeness	Verifies whether the system covers all designated tasks without omission: profile registration, document uploading, and automated eligibility matching.
	Functional Correctness	Assesses whether the inference engine executes the Exclusion Flag Hierarchy without generating false positives or incorrect scholarship recommendations.
	Functional Appropriateness	Examines whether the centralized dashboard facilitates rapid task completion within the AdDU OSA operational context.
Performance Efficiency	Time Behavior	Measures system response times, document ingestion speed, and the query latency of the dynamic faceted search interface.
	Resource Utilization	Monitors cloud resource optimization, ensuring compound indexing minimizes CPU and database load during peak student traffic.

Evaluation will be conducted through task-based usability testing with a purposive sample of 10 to 15 participants, complemented by two sequential post-session instruments: the standardized 10-item System Usability Scale (SUS), which measures subjective perceived usability and addresses Research Question 4, and the Post-Prototype Perceived Effort Survey (Appendix C), which enables direct item-level comparison against the pre-study baseline established in Appendix B to address Research Questions 1 through 3. By systematically measuring error rates and time-on-task against the ISO/IEC 25010 Functional Suitability and Performance Efficiency dimensions, and by comparing pre- and post-prototype perceived effort scores for scholarship discovery, eligibility matching, document management, and deadline tracking, ScholarPath AdDU will be empirically validated as a functional, efficient, and user-centric solution to AdDU's administrative bottlenecks.

References

- [1] Aboitiz Foundation. 2024. Aboitiz Future Leaders Scholarship Program. Retrieved from [Aboitiz Foundation | Aboitiz Future Leaders Scholarship](#)
- [2] AdDU Office of Student Affairs (OSA). 2024. Internal records and scholarship program guidelines. Ateneo de Davao University. Retrieved from [Scholarships and Financial Aid for College | Ateneo de Davao University](#)
- [3] Al-Ayyubi, M. S., & Maulana, B. 2023. Designing a web-based information system for scholarship management: Supporting access and rapid dissemination of information. *Information Technology Engineering Journals (ITEJ)* 8, 1 (2023), 14-33. Retrieved from [Designing a Web-Based Information System for Scholarship Management: Supporting Access and Rapid Dissemination of Information | ITEJ \(Information Technology Engineering Journals\)](#)
- [4] Mahdi, M. N., Ahmad, A. R., Natiq, H., Subhi, M. A., and Qassim, Q. S. 2021. Comprehensive Review and Future Research Directions on Dynamic Faceted Search. *Applied Sciences* 11, 17 (2021), 8113. DOI: [Comprehensive Review and Future Research Directions on Dynamic Faceted Search](#)
- [5] Amer, M. 2025. A Smart Scholarship Management System: All-in-one Solution. Capstone project, Lebanese American University. LAUR Repository. Retrieved from [A Smart Scholarship Management System: All-in-one Solution](#)
- [6] Ateneo de Davao University. n.d. A-1 scholarship program. Retrieved from [A-1 Scholarship Program | Ateneo de Davao University](#)
- [7] Ateneo de Davao University. n.d. Financial aid FAQs. Retrieved from [Financial Aid FAQs | Ateneo de Davao University](#)
- [8] Ateneo de Davao University. n.d. List of scholarship grants. Retrieved from [List of Scholarship Grants | Ateneo de Davao University](#)
- [9] Ateneo de Davao University. n.d. Scholarships and financial aid for college. Retrieved from [Scholarships and Financial Aid for College | Ateneo de Davao University](#)

- [10] Ateneo de Davao University. n.d. Student assistants. Retrieved from [Student Assistants | Ateneo de Davao University](#)
- [11] Atenews. 2024. Tuition hike set for next AY spurs mixed reactions from AdDU community. Atenews. Retrieved from [Tuition hike set for next AY spurs mixed reactions from AdDU community](#)
- [12] Bagaporo, M. D. L., & Espinosa, M. L. n.d. Online scholarship management system for Camarines Sur Polytechnic Colleges. *Global Scientific Journal*. Retrieved from [online scholarship management system for camarines sur polytechnic colleges](#)
- [13] Bangor, A., Kortum, P., & Miller, J. 2008. An empirical evaluation of the System Usability Scale. *International Journal of Human-Computer Interaction* 24, 6 (2008), 574-594. DOI: [Full article: An Empirical Evaluation of the System Usability Scale](#)
- [14] Marthea Andrea O. Daluyon and Ronel J. Bilog. 2025. Scholarship management information system for state university and colleges in Rizal. *Journal of Innovative Technology Convergence* 7, 2 (2025). Retrieved from [Scholarship Management Information System using Machine Learning Models](#)
- [15] Buchanan, B. G., & Duda, R. O. 1983. Principles of rule-based expert systems. *Advances in Computers* 22 (1983), 163-216.
- [16] Catibog, V. 2014. Funds flow and financial control analysis of the Student Financial Assistance Programs (StuFAP). Discussion Paper No. 2014-13. Philippine Institute for Development Studies. Retrieved from [Funds Flow and Financial Control Analysis of the Student Financial Assistance Programs \(StuFAP\)](#)
- [17] Codd, E. F. 1970. A relational model of data for large shared data banks. *Communications of the ACM* 13, 6 (1970), 377-387. DOI: [A relational model of data for large shared data banks | Communications of the ACM](#)
- [18] Commission on Higher Education (CHED). 2013. Guidelines on student affairs and services (CHED Memorandum Order No. 09, s. 2013).
- [19] Coordinating Council of Private Educational Associations of the Philippines (COCOPEA). 2024. Private schools seeking tuition hikes. COCOPEA.
- [20] DeLone, W. H., & McLean, E. R. 2003. The DeLone and McLean model of information systems success: A ten-year update. *Journal of Management Information Systems* 19, 4 (2003), 9-30. DOI: [The DeLone and McLean Model of Information Systems Success: A Ten-Year Update](#)
- [21] Department of Science and Technology Science Education Institute (DOST-SEI). 2024. 2025 DOST undergraduate scholarships now open. DOST-SEI.

- [22] Distor, C. B., & Khaltar, O. 2022. What motivates local governments to be efficient? Evidence from Philippine cities. *Sustainability* 14, 15 (2022), 9426. DOI: [What Motivates Local Governments to Be Efficient? Evidence from Philippine Cities](#)
- [23] Dix, A., Finlay, J., Abowd, G., & Beale, R. 2004. *Human-computer interaction* (3rd ed.). Pearson Education. Retrieved from [Human Computer Interaction](#)
- [24] Ferraiolo, D. F., & Kuhn, D. R. 1992. Role-based access controls. In *Proceedings of the 15th NIST-NCSC National Computer Security Conference*, 554-563. Retrieved from [Role-Based Access Controls](#)
- [25] Glushko, R. J. (Ed.). 2016. *The discipline of organizing* (4th ed.). MIT Press.
- [26] Halili, R. 2014. Assessment of student financial assistance programs (StuFAP) policies, procedures, and control mechanisms. Discussion Paper No. 2014-09. Philippine Institute for Development Studies. Retrieved from [Assessment of Student Financial Assistance Programs \(StuFAP\) Policies, Procedures, and Control Mechanisms](#)
- [27] Hoda, R., Salleh, N., & Grundy, J. 2018. The rise and evolution of agile software development. *IEEE Software* 35, 5 (2018), 58-63. DOI: [The Rise and Evolution of Agile Software Development | IEEE Journals & Magazine](#)
- [28] IBM. 2024. What is event-driven architecture? IBM Think. Retrieved from [What Is Event-Driven Architecture? | IBM](#)
- [29] IMARC Group. 2024. Philippines higher education market size 2034. IMARC Industry Reports. Retrieved from [Philippines Higher Education Market Size 2034](#)
- [30] Indraj, C., & Dominic, J. 2024. Remote Access through Library Automation in Academic Libraries. *Journal of Information Technology and Sciences* 10, 2 (2024), 25–36.(2024).
- [31] ISO/IEC. 2023. ISO/IEC 25010:2023 Systems and software engineering - Systems and software quality requirements and evaluation (SQuaRE) Product quality model. International Organization for Standardization.
- [32] Kent, W. 1983. A simple guide to five normal forms in relational database theory. *Communications of the ACM* 26, 2 (1983), 120-125. Retrieved from [A simple guide to five normal forms in relational database theory | Communications of the ACM](#)
- [33] Linnes, C., Ronzoni, G., Agrusa, J., & Lema, J. 2022. Emergency remote education and its impact on higher education: A temporary or permanent shift in instruction? *Education Sciences* 12, 10 (2022), 721. DOI: [Emergency Remote Education and Its Impact on Higher Education: A Temporary or Permanent Shift in Instruction?](#)

- [34] Mell, P., & Grance, T. 2011. The NIST definition of cloud computing. NIST Special Publication 800-145. National Institute of Standards and Technology. DOI: <https://doi.org/10.6028/NIST.SP.800-145>
- [35] NIST. 2004. Role-based access control. Computer Security Resource Center. Retrieved from [Role Based Access Control | CSRC](#)
- [36] Orgianus, Y., Lapalanti, F. R. I., Tarmizi, H., & Oemar, H. 2024. Optimizing scholarship distribution: A management information system approach. *Acta Logistica* 11, 2 (2024), 197-209. Retrieved from [Optimizing scholarship distribution: a management information system approach](#)
- [37] Pirolli, P., & Card, S. 1999. Information foraging. *Psychological Review* 106, 4 (1999), 643-675. DOI: [Information foraging](#).
- [38] Sandhu, R. S., Coyne, E. J., Feinstein, H. L., & Youman, C. E. 1996. Role-based access control models. *IEEE Computer* 29, 2 (1996), 38-47. DOI: [Role-based access control models](#)
- [39] Schwaber, K., & Beedle, M. 2001. *Agile software development with Scrum*. Prentice Hall.
- [40] SM Foundation. 2024. SM College Scholarship Program. SM Foundation Inc.
- [41] Tanenbaum, A. S., & Van Steen, M. 2007. *Distributed systems: Principles and paradigms* (2nd ed.). Prentice Hall. Retrieved from [Distributed Systems 2nd edition \(2007\)](#)
- [42] Tullis, T. S., & Stetson, J. N. 2004. A comparison of questionnaires for assessing website usability. In *Usability Professionals' Association Annual Conference* (Minneapolis, MN). Retrieved from [A Comparison of Questionnaires for Assessing Website Usability](#)
- [43] Tunkelang, D. 2009. *Faceted Search*. Synthesis Lectures on Information Concepts, Retrieval, and Services. Morgan & Claypool Publishers.
- [44] Unified Student Financial Assistance System for Tertiary Education (UniFAST). 2025. Official launching of e-CAV.
- [45] U.S. Embassy in the Philippines. 2024. U.S. launches Php1.6-billion program to boost Philippine higher education. Retrieved from [U.S. Launches Php1.6-Billion Program to Boost Philippine Higher Education](#)
- [46] Kayanja, W., Kyambade, M., and Kiggundu, T. 2025. Exploring digital transformation in higher education setting: the shift to fully automated and paperless systems. *Cogent Education* 12, 1 (2025), Article 2489800. DOI: [Exploring digital transformation in higher education setting: the shift to fully automated and paperless systems](#)
- [47] Vandana, Kumar, S., & Kumar, V. 2024. Examining the impact of online service convenience on user engagement and intention to continuously use e-resources: mediating role of attitude. *Global*

Knowledge, Memory and Communication. DOI: [Examining the impact of online service convenience on user engagement and intention to continuously use e-resources: mediating role of attitude](#)

[48] World Bank. 2022. Digital transformation of Philippine higher education. Report No. AUS0002964. World Bank Publications. Retrieved from [Digital Transformation of Philippine Higher Education](#)

[49] Yoo, S. J., & Huang, W. D. 2011. Comparison of Web 2.0 technology acceptance level based on cultural differences. *Educational Technology & Society* 14, 4 (2011), 241-252

[50] Kanwal, B., Shoukat, R. S., Rehman, S. U., Kundi, M., AlSaedi, T., & Alahmadi, A. 2024. A new framework for scholarship predictor using a machine learning approach. *Intelligent Automation & Soft Computing*, 39(5), 829–854. DOI: <https://doi.org/10.32604/iasc.2024.054645>

Appendix A: Financial Aid Pipelines Listed at AddU.

#	Scholarship Name	Category	Source URL / Reference
1	CHED-Regional Scholarship Program	State-Sponsored	https://www.addu.edu.ph/list-of-scholarship-grants/
2	CHED-National Scholarship Program	State-Sponsored	https://www.addu.edu.ph/list-of-scholarship-grants/
3	CHED-CSSG (Special Study Grant)	State-Sponsored	https://www.addu.edu.ph/list-of-scholarship-grants/
4	Private Education Student Financial Assistance Program (PESFA)	State-Sponsored	https://www.addu.edu.ph/list-of-scholarship-grants/
5	State Scholarship Program (SSP)	State-Sponsored	https://www.addu.edu.ph/list-of-scholarship-grants/
6	National Integration Study Grant Program (NISGP)	State-Sponsored	https://www.addu.edu.ph/list-of-scholarship-grants/
7	Selected Ethnic Group Educational Assistance Program (SEGEAP)	State-Sponsored	https://www.addu.edu.ph/list-of-scholarship-grants/
8	DOST-SEI Merit Scholarship Program	State-Sponsored	https://www.addu.edu.ph/list-of-scholarship-grants/

9	DOST RA7687 Science and Technology Scholarship	State-Sponsored	https://www.addu.edu.ph/list-of-scholarship-grants/
10	DOST Junior Level Assistance Program (JLAP)	State-Sponsored	https://www.addu.edu.ph/list-of-scholarship-grants/
11	Government Service Insurance System (GSIS) Scholarship Fund	State-Sponsored	https://www.addu.edu.ph/list-of-scholarship-grants/
12	AFP Educational Benefit System Office (AFPEBSO)	State-Sponsored	https://www.addu.edu.ph/list-of-scholarship-grants/
13	US Department of Veterans Affairs Education Benefits	State-Sponsored	https://www.addu.edu.ph/list-of-scholarship-grants/
14	Antonio A. Villanueva Endowment Fund	Internally Funded	https://www.addu.edu.ph/list-of-scholarship-grants/
15	Ateneo Alumni Association of Canada	Internally Funded	https://www.addu.edu.ph/list-of-scholarship-grants/
16	Ateneo Alumni Association of Southern California	Internally Funded	https://www.addu.edu.ph/list-of-scholarship-grants/
17	Alfonso Frances Marie Ybanez	Internally Funded	https://www.addu.edu.ph/list-of-scholarship-grants/
18	Arts and Sciences Fund	Internally Funded	https://www.addu.edu.ph/list-of-scholarship-grants/
19	Rev. Fr. John Dotterweich SJ Scholarship Fund	Internally Funded	https://www.addu.edu.ph/list-of-scholarship-grants/
20	Rev. Fr. Rodolfo Malasmas SJ Scholarship Fund	Internally Funded	https://www.addu.edu.ph/list-of-scholarship-grants/
21	Bank of Philippine Islands (BPI) Foundation	Internally Funded	https://www.addu.edu.ph/list-of-scholarship-grants/
22	De La Paz Scholarship Fund	Internally Funded	https://www.addu.edu.ph/list-of-scholarship-grants/
23	Davao Jesuit Memorial Scholarship Fund	Internally Funded	https://www.addu.edu.ph/list-of-scholarship-grants/
24	Rev. Fr. Edmundo M. Martinez, SJ Endowment Fund	Internally Funded	https://www.addu.edu.ph/list-of-scholarship-grants/
25	Grant-In-Aid (GIA) Fund	Internally Funded	https://www.addu.edu.ph/scholarships-and-financial-aid-for-college/
26	Kalasag Scholarship Fund	Internally Funded	https://www.addu.edu.ph/list-of-scholarship-grants/
27	Jubilee Scholarship Fund (Valedictorian &	Internally Funded	https://www.addu.edu.ph/list-of-scholarship-grants/

	Salutatorian)		
28	Montemayor Scholarship Fund	Internally Funded	https://www.addu.edu.ph/list-of-scholarship-grants/
29	St. Aloysius Gonzaga Scholarship Fund	Internally Funded	https://www.addu.edu.ph/list-of-scholarship-grants/
30	Sycip, Gorres, Velayo (SGV) and Company	Internally Funded	https://www.addu.edu.ph/list-of-scholarship-grants/
31	St. John Berchman Scholarship	Internally Funded	https://www.addu.edu.ph/list-of-scholarship-grants/
32	Vicente L. Babao Scholarship Foundation	Internally Funded	https://www.addu.edu.ph/list-of-scholarship-grants/
33	Jose & Avelina Buktaw Scholarship	Internally Funded	https://www.addu.edu.ph/list-of-scholarship-grants/
34	Rogelio Alama Scholarship	Internally Funded	https://www.addu.edu.ph/list-of-scholarship-grants/
35	SC Johnson and Son, Inc. Scholarship Grant	Internally Funded	https://www.addu.edu.ph/list-of-scholarship-grants/
36	A-1 Scholarship Program (Micro-Philanthropy)	Internally Funded	https://www.addu.edu.ph/a-1-scholarship-program/
37	Alfonso Yuchenco (AY) Foundation, Inc.	Corporate / External	https://www.addu.edu.ph/list-of-scholarship-grants/
38	Antonio O. Floirendo Sr., Foundation, Inc.	Corporate / External	https://www.addu.edu.ph/list-of-scholarship-grants/
39	Archimedes and Samad Lu Foundation Scholarship Grant	Corporate / External	https://www.addu.edu.ph/list-of-scholarship-grants/
40	Ateneo Alumni Association of British Columbia	Corporate / External	https://www.addu.edu.ph/list-of-scholarship-grants/
41	Ben and Quennie Balaba Scholarship Grant	Corporate / External	https://www.addu.edu.ph/list-of-scholarship-grants/
42	Carmudi Philippines Scholarship Program	Corporate / External	https://www.addu.edu.ph/list-of-scholarship-grants/
43	Equitable PCIBank Foundation	Corporate / External	https://www.addu.edu.ph/list-of-scholarship-grants/
44	EMCOR, Inc.	Corporate / External	https://www.addu.edu.ph/list-of-scholarship-grants/
45	Jesus V. del Rosario (JVR) Foundation Scholarship	Corporate / External	https://www.addu.edu.ph/list-of-scholarship-grants/
46	Jollibee Foundation, Inc.	Corporate / External	https://www.addu.edu.ph/list-of-scholarship-grants/

47	Metrobank Foundation Scholarship Fund	Corporate / External	https://www.addu.edu.ph/list-of-scholarship-grants/
48	Nelly Kellog Van Shaik (NKVS) Scholarship	Corporate / External	https://www.addu.edu.ph/list-of-scholarship-grants/
49	PAGCOR Scholarship Fund	Corporate / External	https://www.addu.edu.ph/list-of-scholarship-grants/
50	Shoemart (SM) Foundation, Inc.	Corporate / External	https://www.addu.edu.ph/list-of-scholarship-grants/
51	Vicente B. Bello Scholarship	Corporate / External	https://www.addu.edu.ph/list-of-scholarship-grants/
52	Tanging Yaman Foundation	Corporate / External	https://www.addu.edu.ph/list-of-scholarship-grants/
53	Davao Light / Aboitiz Foundation (MICD Scholarship)	Corporate / External	https://aboitizpower.com/news/corporate-social-responsibility/davao-light-aboitiz-foundation-continue-support-indigenous-scholars
54	Student Assistant (SA) Program	Specialized / Service	https://www.addu.edu.ph/student-assistants/

Appendix B: Pre-Study Student Problem Validation Survey

Question 1: Have you ever applied for a scholarship or financial aid at Ateneo de Davao University using the manual/traditional application process?

- **Question Type:** Multiple Choice
- **Options:**
 - Yes (*Continue to next section*)
 - No (*Submit form - they are not your target respondents*)

Component 1 - Perceived Time-on-Process

Please recall your most recent experience applying for a scholarship at AdDU using the traditional, paper-based process.

Question 2: How much total time did you spend navigating the entire manual application process? (*This includes the time spent discovering the scholarship, gathering requirements, routing forms like Form 230-SCH, and physically submitting the documents*).

- **Question Type:** Multiple Choice

- **Options:**
 - Less than 1 day
 - 1–3 days
 - 4–7 days
 - 1–2 weeks
 - More than 2 weeks
-

Component 2 - Perceived Effort Scale

The following questions assess the difficulty of specific tasks during the manual application process.

Question 3: On a scale of 1 to 5, please rate how easy or difficult it was to complete the following application steps:

- Question Type: Likert Grid (5-point scale)
 - Scale Columns:
 - 1 - Very Easy
 - 2 - Easy
 - 3 - Neutral
 - 4 - Difficult
 - 5 - Very Difficult
 - Row Sub-tasks:
 1. Finding out which specific scholarships or grants you were eligible for.
 2. Determining whether you specifically met all eligibility criteria — such as QPI requirement, household income ceiling, and degree program restriction — for a scholarship before submitting an application.
 3. Gathering, printing, and preparing the required physical documents.
 4. Physically submitting the completed application package to the respective offices.
 5. Tracking the status of your application after submission.
 6. Keeping track of multiple scholarship deadlines simultaneously throughout the application period.
-

Component 3: Application Accuracy and Deadline History

This section helps establish how often errors and missed deadlines occurred under the manual process.

Question 4: Have you ever submitted a scholarship application and later discovered that you did not meet the eligibility requirements for that scholarship?

- Question Type: Multiple Choice

- Options:
 - Yes
 - No
 - I am not sure

Question 5: *(Displayed only if "Yes" is selected in Question 4.)* Approximately how many times did this occur?

- Question Type: Multiple Choice
- Options:
 - Once
 - 2–3 times
 - 4 or more times

Question 6: Have you ever missed a scholarship application deadline while using the manual application process?

- Question Type: Multiple Choice
- Options:
 - Yes
 - No

Question 7: *(Displayed only if "Yes" is selected in Question 6.)* What was the primary reason you missed that deadline?

- Question Type: Multiple Choice
- Options:
 - I was unaware that the deadline existed or had passed
 - I did not have enough time to prepare the required physical documents
 - I forgot to follow up on the application after initial discovery
 - Other *(Please specify)*

Appendix C: Post-Prototype Perceived Effort Survey

Component 1: Perceived Time-on-Process

Based on your experience using the ScholarPath AdDU prototype during the testing session, please answer the following.

Question 1: How much total time do you estimate the digital scholarship application process through ScholarPath AdDU would require? *(This includes the time spent discovering a scholarship, verifying your eligibility, uploading documents to the Document Vault, and submitting an application.)*

- Question Type: Multiple Choice
- Options:
 - Less than 1 day
 - 1–3 days
 - 4–7 days
 - 1–2 weeks
 - More than 2 weeks

Component 2: Perceived Effort Scale

The following questions assess the difficulty of specific tasks while using ScholarPath AdDU. Please use the same scale you applied in the pre-study survey for direct comparison.

Question 2: On a scale of 1 to 5, please rate how easy or difficult it was to complete each of the following steps using ScholarPath AdDU:

- Question Type: Likert Grid (5-point scale)
- Scale Columns:
 - 1 — Very Easy
 - 2 — Easy
 - 3 — Neutral
 - 4 — Difficult
 - 5 — Very Difficult
- Row Sub-tasks:
 1. Finding out which specific scholarships or grants you were eligible for using the centralized scholarship repository.
 2. Determining whether you specifically met all eligibility criteria using the results produced by the Smart Eligibility Checker.
 3. Uploading and managing the required documents through the Document Vault.
 4. Submitting a completed application digitally through the platform.
 5. Tracking the status of your application through the student dashboard.
 6. Keeping track of multiple scholarship deadlines using the automated notification system.

Component 3: System Feature Effectiveness

This section assesses the perceived effectiveness of ScholarPath AdDU's core features in addressing specific problems identified in the manual process.

Question 3: Do you believe the Smart Eligibility Checker would effectively prevent you from applying to scholarships for which you do not qualify?

- Question Type: Multiple Choice
- Options:
 - Yes, I believe it would prevent mismatched applications
 - Partially — it would help, but I might still need to verify manually
 - No, I do not think it would reliably prevent ineligible applications
 - Unsure

Question 4: To what extent do you believe the automated deadline notification and alert system in ScholarPath AdDU would make it easier to stay aware of upcoming application deadlines?

- Question Type: Likert Scale
- Options:
 - 5 — Strongly Agree
 - 4 — Agree
 - 3 — Neutral
 - 2 — Disagree
 - 1 — Strongly Disagree

Component 4: Overall Comparative Perception

Question 5: Compared to the traditional manual scholarship application process, how would you characterize the overall effort required to complete a scholarship application using ScholarPath AdDU?

- Question Type: Multiple Choice
- Options:
 - Much Easier
 - Somewhat Easier
 - About the Same
 - Somewhat Harder
 - Much Harder

Question 6 (Optional): Which specific feature of ScholarPath AdDU most reduced the difficulty of the scholarship application process for you? Please describe briefly.

- Question Type: Short Answer (Open-ended)